

TARGETING DISASTER AID: A STRUCTURAL EVALUATION OF A LARGE EARTHQUAKE RECONSTRUCTION PROGRAM

MATTHEW D. GORDON

Department of Economics, Paris School of Economics

YUKIKO HASHIDA

Agricultural and Applied Economics, University of Georgia

ELI P. FENICHEL

School of the Environment, Yale University

This paper studies the question of how to target aid after a natural disaster. Disaster aid programs often use property damage as a criterion for eligibility. A household's ability to insure against shocks may be harder to observe but more important in determining how the disaster affects welfare. We develop a model of demand for reconstruction aid and estimate the model parameters using data from a household survey following the 2015 Nepal earthquake and moments derived from a spatial regression discontinuity design. We use the model to estimate welfare from counterfactual targeting strategies. Conditioning aid on damages increases welfare by 9% relative to a random allocation, but a geographically targeted aid strategy based on earthquake intensity would have increased welfare by 12%. Furthermore, policymakers face trade-offs between targeting strategies aimed at consumption smoothing and those prioritizing poor households. Damage-based targeting performed worse than random by a poverty-focused measure of welfare.

KEYWORDS: Natural Disasters, Targeting, Aid, Consumption Smoothing.

MG: matthew.gordon@psemail.eu; YH: yhashida@uga.edu; EF: eli.fenichel@yale.edu. Thanks to Ken Gillingham, Matthew Kotchen, Mushfiq Mobarak, Mark Rosenzweig, Marc Conte, Geoff Barrows, Stephane Hallegatte, Subhrendu Pattanayak, Ivan Rudik, Klaas Van't Veld, Thomas Walker, Michael Peters, Stephen Newbold, Luke Sanford, Ethan Addicott, Stephanie Weber, Simon Lang, Eugene Tan, Emmanuel Murray Leclair, Bishal Kumar Chalise, Nirmal Kumar Raut, Santiago Saavedra, Robert Gonzalez, Mark Buntaine, Karen Seto, and the participants of LSE Environment Week, NEUDC and seminars at UCL, TSE, Exeter, and the University of Wyoming for feedback. Support for this research was provided by the Molly Macauley Award from Resources for the Future. MG, YH, and EF conceived of the broad ideas. MG developed the theory and empirical strategies, performed all analyses, and wrote the paper. MG, YH, and EF reviewed the paper.

1. INTRODUCTION

Cash transfers to households after a natural disaster could be an effective way to mitigate the negative impacts of the disaster and promote reconstruction, especially when households lack insurance or other means of smoothing the shock. A common approach to targeting aid in post-disaster reconstruction programs is to make eligibility conditional on disaster-related property damage. The relevance of using property damage as a proxy for need is not well established in economic theory or empirically, however. Other forms of vulnerability, including barriers to credit and insurance markets, are harder to observe, but they are important determinants of a household's post-disaster welfare and ability to recover.

Yet property damage may be an attractive criterion for targeting because it is a highly visible mark of the disaster's impacts. In the US, FEMA spends on average more than \$1 billion annually on aid to homeowners who suffer housing damages in a disaster. In disasters in low-income countries, aid allocated on the basis of property damages is often the only source of cash aid available to households.¹

This paper studies the question of how aid should be targeted after a disaster, using the 2015 Nepal earthquake as a case study. The 2015 earthquake was the largest in Nepal since 1934, and by some estimates left 12% of the country homeless ([The Asia Foundation, 2016b](#)). In the months after the earthquake, households could qualify for a large cash grant based on the degree of damage to their house. We develop a framework to measure the effects of this aid on welfare and reconstruction, taking into account different possible objectives.

We analyze counterfactual targeting policies by estimating a structural model of household behavior using a panel of rural households covering the reconstruction period. The model assumes that households can draw upon savings and informal insurance to smooth shocks, but they face a borrowing constraint. We build evidence for these assumptions by exploiting a spatial discontinuity in aid delivery. Regression discontinuity estimates show that aid increases consumption and housing investment, but decreases remittances – a com-

¹FEMA numbers based on 2010-2020 average from [FEMA \(2022\)](#). See Table [B.I](#) in the Appendix for descriptions of recent earthquakes in low-income countries and associated aid programs.

mon form of informal insurance. We use these estimates to calibrate the parameters of the structural model.

In the context of anti-poverty programs, targeting strategies are often evaluated according to a utilitarian social welfare function, which assesses how accurately the program directs aid to the households with the highest marginal value for cash (Niehaus et al., 2013, Hallegatte et al., 2016, Hanna and Karlan, 2017, Hanna and Olken, 2018, Aiken et al., 2022, Banerjee et al., 2023). We find that targeting based on property damages is not effective according to this objective – it generates just 44% of the welfare gains of a random allocation.

On the other hand, the objectives of disaster aid are likely different – policymakers might seek to help recipients recover from a shock, regardless of how well-off they were initially. We derive a measure of household ‘demand for aid’, that can be interpreted as a household’s willingness-to-pay to smooth a shock. We measure demand as the amount of income a household would be willing to forgo in the future in exchange for receiving aid in the present. Demand for aid goes down when households have alternative means of insurance, and increases when liquidity constraints bind. This measure distinguishes a household’s willingness-to-pay from its ability-to-pay, which is relevant when households face credit market failures. Household estimates of demand for aid derived from our structural model correlate with costly coping strategies, including taking on high-interest loans, cutting food consumption, and children working.

According to this measure of welfare, the damage-based targeting strategy pursued in Nepal increased welfare by 9% relative to a random allocation of aid. Other simple strategies perform as well, or better, however. A strategy that targeted all households in villages with the highest earthquake intensity increases welfare by 12%, and a universal allocation, that gives a smaller amount of money to all households in the affected districts, increases welfare by 11%, without accounting for administrative cost savings or delays.

These alternatives perform well for two reasons. First, some households that experienced damages have ‘insurance’ – access to loans and remittances that allow them to recover regardless of whether they receive aid. Second, some households have no property damage, but they still have high value for aid in order to meet their immediate need for consumption. This is particularly true in villages that experienced heavy damages, as local labor markets were disrupted, and households faced different types of shocks. These findings suggest that

unconditional aid, allocated according to simple geographic rules, can outperform costly and time-consuming damage-based allocations.

The superiority of the village-based targeting program is robust to alternative parameters and assumptions about household income dynamics, although the relative benefits of universal aid are sensitive to estimates of the curvature of household value functions. We also study how the benefits of targeting depend on the distribution of liquid and housing wealth. Targeting becomes more valuable as inequality increases, while universal aid performs better when households are relatively homogeneous.

In addition, we explore objective functions that assume the social planner wants to increase either aggregate consumption or housing reconstruction. This latter goal could be important if housing reconstruction has large externalities or general equilibrium effects. We find that, while damage-based targeting does better than the random allocation in promoting reconstruction, it does worse than targeting households with low liquid wealth, as many poor households live in substandard housing even if they didn't experience earthquake damages.

This paper contributes to the growing literature on targeting aid by developing a framework for studying the targeting of disaster aid. The existence of nepotism and politics in the distribution of disaster aid has been well documented ([Basurto, Dupas and Robinson, 2020](#), [Tarquinio, 2022](#), [Mahadevan and Shenoy, 2023](#)), including in the context of the Nepal earthquake ([Bhusal et al., 2022](#), [Pathak and Schündeln, 2022](#)). Our normative analysis critically examines the objectives of disaster aid, and the results show that even well-intentioned targeting strategies may be suboptimal.

We build on previous work using structural models to study the welfare implications of disaster aid. To date this literature has mainly focused on the US and the Hurricane Katrina recovery. [Gregory \(2017\)](#) and [Fu and Gregory \(2019\)](#) study program design to minimize moral hazard and maximize rebuilding externalities—important concerns in urban contexts and wealthy countries with more comprehensive social insurance programs. In contrast, we adapt the incomplete insurance framework of [Kaboski and Townsend \(2011\)](#) to emphasize rural income risk and the welfare gains from targeting—first-order concerns in the design of social insurance in poor countries .

In addition, we show empirical evidence on the causal effects of disaster aid on consumption, income, investment, and informal insurance using a spatial regression disconti-

nunity approach. The research design relies on the fact that households in certain districts were prioritized for aid, and that households close to the borders of those districts experienced similar levels of earthquake damages, but differed in their likelihood of receiving aid. The results show that receipt of aid has large effects on household consumption and housing reconstruction. Aid also substituted for some types of informal insurance including remittances and migration. These effects are consistent with our model of partially insured households.

We use Townsend-style regressions to show evidence that there is considerable heterogeneity in households' ability to smooth consumption ([Townsend, 1994](#)). The forms of insurance used in this context – mainly loans, remittances and migration – are not equally available to all households. Wealthy households are less likely to experience damages from the disaster, and they are also better able to recover by drawing on sources of informal insurance. This echoes previous work finding that the negative effects of disasters on GDP are amplified in poorer settings (see [Kellenberg and Mobarak 2011](#); [Dell, Jones and Olken 2014](#); and [Botzen, Deschenes and Sanders 2019](#) for reviews).

In the US, studies show transitory impacts on income for even very large disasters, though recovery is aided by reliance on insurance and formal social safety nets ([Deryugina, 2017](#), [Deryugina, Kawano and Levitt, 2018](#), [Gallagher and Hartley, 2017](#), [Billings, Gallagher and Ricketts, 2022](#), [Gallagher, Hartley and Rohlin, 2023](#)). Since formal safety nets are sparse in many low-income countries, heterogeneous access to informal insurance makes targeting aid important. [Tarquinio \(2022\)](#) finds that drought declarations in India increase household consumption, but the effect is smaller when the declarations are poorly targeted.

Our results show substantial heterogeneity in household demand for aid, even between rural households in a low-income country. Furthermore, the heterogeneity that we estimate is correlated with the same characteristics that predict differential ability to smooth consumption. Earthquake-related housing damages increase demand for aid, but a targeting strategy exclusively focused on that proxy will miss households facing other types of vulnerabilities. If these vulnerabilities are not easily observable, geographic targeting, or giving a smaller amount of aid to a larger number of people may be simple policies that improve welfare.

The remainder of the paper is as follows. Section 2 gives context on the earthquake and the aid program, and shows descriptive evidence on household consumption smoothing and informal insurance. Section 3 describes our normative framework and develops the structural model which allows us to estimate household demand for aid. Section 4 estimates the model using the method of simulated moments, including moments derived from a spatial regression discontinuity in the allocation of aid. We then validate the model by comparing the correlates of demand for aid to heterogeneity in consumption smoothing. Section 5 then uses the model to analyze the implications of counterfactual targeting strategies on welfare and reconstruction.

2. CONTEXT AND DATA

With a 2015 per capita GDP of \$3,330, Nepal is one of the poorest countries in the world outside of Africa. Two-thirds of households are employed in agriculture or livestock rearing,² and the economy remains heavily dependent on migrant labor and remittances, which also provide an important means of insuring against environmental shocks.

The Himalayan portion of the country is a subduction zone that experiences frequent earthquakes. In April 2015, the Gorkha earthquake, named after the district where it occurred, was measured at a 7.8 magnitude – the largest earthquake in Nepal since 1934. The earthquake triggered landslides throughout the region, flattening entire villages. The final death toll was nearly 9,000 people, and 12% of the country saw their homes destroyed ([The Asia Foundation, 2016b](#)).

The earthquake took place in a context of poor state capacity. Nepal endured a decade of civil war from 1996-2006, followed by eight years of transitional governments in the lead-up to the earthquake. Riots and a blockade of the border with India, and a drought in 2016, compounded food insecurity in the aftermath of the disaster ([Randell et al., 2021](#)).

In the fourteen most severely affected districts, several small emergency cash grants were distributed within six months of the earthquake. Households that had their homes completely destroyed typically received 25,000 NPR (about \$250) to buy emergency supplies and procure shelter before winter set in. Village Development Committee (VDC) leaders

²Data from World Bank. In 2015 exchange rates averaged close to 100 NPR to \$1.

compiled initial lists of households eligible for benefits, with some discretion in their ability to do so ([The Asia Foundation, 2016b](#), [Pathak and Schündeln, 2022](#)).

An initial needs assessment conducted after the earthquake identified rural housing reconstruction as the largest need area ([Government of Nepal National Planning Commission, 2015](#)). A multilateral donor fund pledged \$4.1 billion for reconstruction, with aid initially targeted at the fourteen most affected districts ([Nepal Earthquake Housing Reconstruction Program Multi Donor Trust Fund, 2016](#)).

Households qualified for reconstruction aid if their house required complete rebuilding, as certified by teams of engineers sent by the central government. This resulted in new beneficiary lists that were, in many cases, less generous than the initial lists, leading to more protests and unrest in some districts. Although some households had started repairs, [The Asia Foundation \(2016a\)](#) found that 75% of displaced households were still living in temporary shelters 18 months after the earthquake, with many more moving back into their partially reconstructed and potentially unsafe homes.

For households that qualified, the reconstruction grants amounted to 300,000 NPR (about \$3,000), delivered in three tranches. The first funds were delivered in early 2016, a full year after the earthquake. By July 2018, 60% of eligible households in the most-affected districts had received at least two installments, compared to only 15% of eligible households outside those districts ([Housing Recovery and Reconstruction Platform, 2018](#)). These reconstruction grants are the focus of the targeting analysis in this paper.

Within villages, earthquake damages negatively correlated with income, assets, and education (see Appendix C). This seemed mainly to reflect wealthier households using better building materials, as cement and reinforced concrete buildings fared better on average. The aid response did not favor the poor, however. [Bhusal et al. \(2022\)](#) find that elite caste members received more aid, even after controlling for assessed damages. This bias was partially reversed in VDCs with mayors from non-elite castes, however.

2.1. Data and Descriptive Statistics

Our primary source of data is the World Bank Household Risk and Vulnerability Survey (WBHRVS) – a representative panel survey of 6,000 rural households from across Nepal that covers 2016-18, with one wave corresponding to each year in that time period ([Walker,](#)

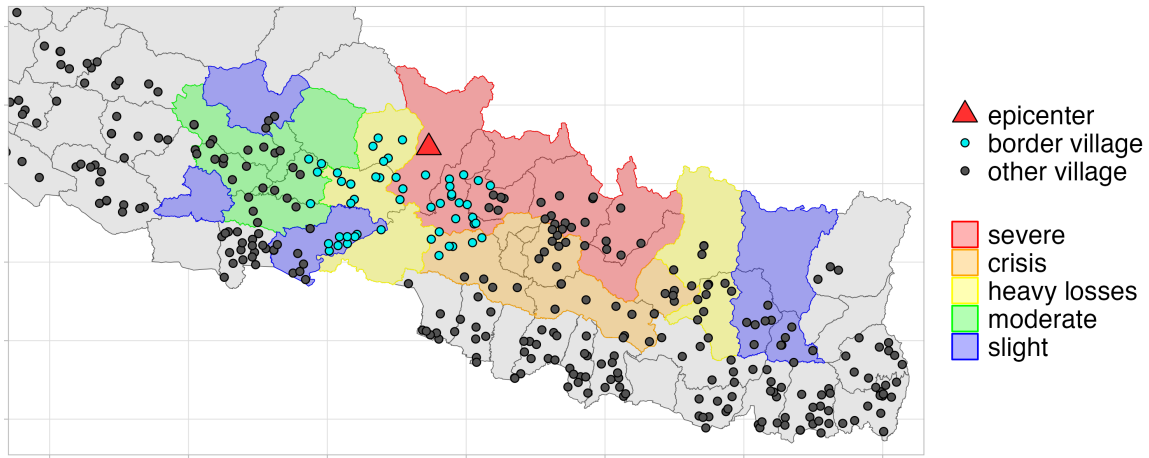


FIGURE 1.—Map of study region. District colors denote earthquake damage designations from [Housing Recovery and Reconstruction Platform \(2018\)](#). ‘Severe’ and ‘Crisis’ districts were prioritized for aid. Black circles represent VDCs present in the survey. Blue circles are VDCs within 59 km of the western border of the ‘Severe’ and ‘Crisis’ districts – the bandwidth selected for inference in the first stage of the regression discontinuity analysis (see Section 4.1).

[Kawasoe and Shrestha, 2019](#)). The survey collected data on household asset, livelihood, and consumption variables, as well as shocks experienced and coping strategies. It also records whether households received earthquake reconstruction aid. The survey was conducted separately from the engineering assessments that determined household eligibility for aid. The locations of sample households are shown in Figure 1. Additionally, we use data from the US Geological Survey to measure slope, elevation, and peak ground acceleration during the earthquake. The latter is used to define a targeting strategy based on villages with the highest earthquake intensity. Finally, to look at the correlates of earthquake damages in Appendix C, we use the post-earthquake building census conducted by the National Reconstruction Authority.

Average income of the households in the sample is 275,000 NPR (Appendix Table B.II). Given an average household size of 4.66, this comes out to be approximately \$1.60 per person per day. This goes up to nearly \$2 when remittances are included. 10% of households report that they skipped a meal in the last 30 days. Long-term migration is an important livelihood strategy. Nearly half of the households in the sample are economically connected to a migrant, with many connected to more than one. More than half of the migrants mentioned by the households in the sample were in another country – with India, Southeast

Asia, and the Middle East as the main destinations. Loans are important in this context as well, with the average household taking 29,000 NPR in short term loans in the past year. The most common source of loans is from friends and relatives (40%) with money lenders and savings groups accounting for most of the rest. Formal banking services account for a negligible fraction of loans.

Of the twenty-one percent of households that report experiencing losses from the earthquake, 97% of those report asset losses, compared to 22% reporting income losses – consistent with findings in the post-disaster needs assessment prioritizing housing as the main focus of reconstruction. In addition to the nature of the earthquake as a shock that primarily destroys fixed capital, another possible explanation for this distribution of losses is that many households' primary source of income comes from migrants overseas, which was unaffected by the earthquake. Thirteen percent of households received aid.

2.2. *Consumption Smoothing*

Targeting could increase the value of aid as a form of social-insurance if it is directed towards households that are least able to smooth consumption ([Chetty, 2006](#)). To test for heterogeneity in household ability to smooth consumption, we use Townsend-style regressions, regressing consumption on income conditional on household and village-year fixed effects ([Townsend, 1994](#)). This tests whether idiosyncratic income shocks correlate with household consumption. Results in Table I reject the null hypothesis of perfect risk sharing. For a 10% reduction in income, households reduce consumption by 1% on average.

There is important heterogeneity across several subgroups in the sample, however. The second column explores heterogeneity by caste and ethnicity. Members of the economically and politically powerful Newar caste smooth consumption more effectively than other groups as indicated by the negative coefficient on the interaction term. This could be consistent with [Bhusal et al. \(2022\)](#) findings on preferential access to aid amongst the politically dominant caste. The last three columns show heterogeneity by sex of household head, value of landholdings, and earthquake exposure. Female headed households and households with above median landholdings smooth consumption more effectively.

Landownership is a proxy for wealth, and the results suggest that wealthier households are better insured. The negative coefficient on female-headed households might be some-

	log(consumption)				
	(1)	(2)	(3)	(4)	(5)
log(income)	0.103*** (0.006)	0.101*** (0.009)	0.102*** (0.006)	0.110*** (0.007)	0.101*** (0.006)
log(income):Dalit		0.009 (0.014)			
log(income):Newar		−0.045** (0.015)			
log(income):Other		0.005 (0.009)			
log(income):Female Head			−0.010*** (0.002)		
log(income):Land > Median				−0.013** (0.006)	
log(income):Quake Affected					0.012 (0.012)
Observations	16 742	16 741	16 742	16 742	16 742
R2	0.792	0.792	0.793	0.792	0.792

TABLE I

CONSUMPTION SMOOTHING

All regressions include household and village-year fixed effects. Standard errors clustered at the survey strata. Households with zero or missing income, or less than 3 observations dropped. Land percentile defined based on household land value during first survey wave. Omitted category for caste is "Brahmin/Chhetri". * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

what surprising in the light of often presumed increased vulnerability amongst these households, however one potential explanation is that 61% of these households receive remittances, compared to 26% of male-headed households, and remittances tend to be strongly counter cyclical, as seen in Table II.A. Notably, earthquake exposed households seem to smooth no worse on average, though this combines the effect of the earthquake with the effect of the aid. Since village-year fixed effects absorb the aggregate shock of the earthquake, we also look at ability to smooth income shocks with respect to caste-year fixed effects as well as using rainfall as an instrument for income, and we find similar patterns of heterogeneity (Appendix Tables B.III and B.IV).

To test which types of transfers are pro- or counter-cyclical with respect to household income fluctuations, we regress various types of transfers on log income, conditional on a household fixed effect. The columns in Table II.A show that remittances, migration, and

A. Within Household Variation

	Earthquake Aid (1)	NGO Transfers (2)	Govt Transfers (3)	Informal Transfers (4)	Cash Savings (5)	Remittances (6)	Migrants (7)	Loans Taken (8)
log(income)	2.370* (1.174)	-0.222 (0.206)	0.001*** (0.000)	0.268 (0.191)	7.606*** (1.884)	-8.678*** (1.660)	-0.051*** (0.012)	-3.557* (1.767)
Observations	16 742	16 742	16 742	16 739	16 671	16 742	16 742	16 728
R2	0.493	0.333	0.650	0.342	0.505	0.604	0.375	0.361
Fixed Effects	Household	Household	Household	Household	Household	Household	Household	Household

B. Within Village-Year Variation

	Earthquake Aid (1)	NGO Transfers (2)	Govt Transfers (3)	Informal Transfers (4)	Cash Savings (5)	Remittances (6)	Migrants (7)	Loans Taken (8)
Land > Median	-0.007 (0.349)	-0.103 (0.179)	0.001 (0.000)	-0.172 (0.728)	34.231*** (3.009)	21.307** (7.287)	0.042** (0.015)	4.052* (1.909)
Observations	16 742	16 742	16 742	16 739	16 671	16 742	16 742	16 728
R2	0.488	0.362	0.121	0.078	0.141	0.140	0.142	0.110
Fixed Effects	Village-Year	Village-Year	Village-Year	Village-Year	Village-Year	Village-Year	Village-Year	Village-Year

TABLE II

TRANSFER INCOME

Panel A shows various forms of transfer income and number of migrants regressed on log labor income conditional on a household fixed effect. Panel B shows the results of a regression of forms of transfer income and number of migrants on an indicator for land value greater than median in the first survey wave conditional on a village-year fixed effect. All dependent variables except migrants in thousands of NPR. Households with zero or missing income or less than 3 observations dropped. Standard errors clustered at the survey strata. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

loans negatively correlate with within-household income fluctuations, suggesting that they are used as forms of informal insurance. Cash savings are pro-cyclical, suggesting that households save in a good year and draw on savings in bad years. Earthquake aid is positively correlated with income shocks, possibly because it arrived more than a year after the event in most cases. Government transfers are also pro-cyclical, highlighting the absence of formal social insurance that responds to shocks.

To test how various forms of transfers correlate with wealth, we regress various forms of transfer income on a dummy for above median landholdings, conditional on a village-year fixed effect that captures aggregate shocks to the village. Table II.B shows that cash savings, remittances, migration, and loans in a given year are positively correlated with household landholdings within villages, suggesting that these sources of support are more easily available to wealthier households.

Taken together these correlations indicate that households vary in their ability to smooth income shocks, possibly due to differential access to savings, loans, remittances and migration opportunities. This implies that aid might have greater benefits if targeted towards

households that lack these forms of informal insurance. In the next section, we present a model of household behavior and a normative framework informed by these results. We then validate our model by confirming that the households characteristics that correlate with demand for aid are similar to the household characteristics associated with limited ability to smooth consumption.

3. TARGETING CRITERIA FOR RECONSTRUCTION AID

Judging the merits of different allocations of aid requires making interpersonal comparisons between households. Assuming the planner has a fixed budget means that giving aid to one household requires not giving it to another. Consider a planner allocating aid to maximize a social welfare function W that aggregates a weighted sum of the indirect utility functions of households indexed by i :

$$W^* = \max_{a_{it}} \sum_{i=0}^N \alpha_i V_{it}(a_{it}, x_{it}, h_{it}, P_{it}) \quad (1)$$

$$\text{such that } \sum_{i=0}^N a_{it} \leq A$$

V_{it} is the indirect utility function of household i at time t that depends on aid received (a_{it}), and other sources of heterogeneity between households. These include the quality of their housing stock, reflecting damage from the earthquake (h_{it}), access to liquidity for reconstruction (x_{it}), and permanent income (P_{it}). Ability to access liquid wealth is a particularly important determinant of self-insurance for a large covariate shock in a developing country where informal networks form an important risk sharing mechanism (Munshi and Rosenzweig, 2016, Morten, 2019). Each household has Pareto-weight α_i , which the planner chooses based on targeting objectives. A is the aid budget.

Much of the targeting literature focused on poverty alleviation has used a utilitarian social welfare function (SWF) to assess benefits, where all households have $\alpha_i = 1$ (Hanna and Olken, 2018). Given a concave value function, this functional form favors redistribution to the poorest households. While this is potentially appropriate for analyzing poverty-relief programs, it may not fit the goals of disaster-reconstruction aid as well. A utilitarian planner would only condition aid on disaster-related damages if those damages were a good proxy

for a high marginal utility of aid. We will analyze to what extent this is true in our setting, but we also examine SWFs that reflect other objectives.

In particular, we consider a SWF consistent with the idea that the purpose of disaster aid is to help households smooth consumption and maintain their quality-of-life in the aftermath of the disaster. Intuitively, this type of objective is more consistent with a targeting strategy based on disaster damages.

Typical analyses of the allocation of non-market goods seek to maximize willingness-to-pay – a metric that makes little conceptual sense when considering cash aid, since the nominal value of the aid is the same to all households. Instead, to implement an objective focused on consumption smoothing, we measure what percentage of future income a household would be willing to give up in exchange for the amount of aid disbursed by the reconstruction program. This value can be inferred by solving for a ‘tax’ on future income, τ_i , such that a household is indifferent between their current liquidity and income, and receiving the aid, but paying a tax on future income:

$$V(x_{it}, h_{it}, P_{it}) = V(x_{it} + a_{it}, h_{it}, (1 - \tau_i)P_{it}). \quad (2)$$

This is the household’s compensating variation for liquidity in terms of future income. A notion of the household’s ‘willingness-to-pay’ (WTP) for aid can be calculated as a discounted sum of expected future tax payments, with discount factor β . In discrete, infinite time:

$$WTP_{it} = \sum_{t=0}^{\infty} \beta^t (\tau_i P_{it}) = \frac{\tau_i P_{it}}{1 - \beta}. \quad (3)$$

Measuring demand for aid in this manner is like asking a household how much they would be willing to borrow from the future in order to smooth the present shock. It is a useful concept in a context with credit market constraints, since if markets were complete, households could borrow at prevailing interest rates. With the borrowing constraint, however, households might be willing to pay above market interest rates (and be able to repay in expectation), but they are unable to find a lender willing to offer those terms. In this way, we can think of a concept of willingness-to-pay that is distinct from ability-to-pay.

This form of SWF satisfies some desirable theoretical properties, including an idealized form of inter-temporal Kaldor-Hicks efficiency. If households were allowed to make transfers through time, households that don't receive aid could be compensated with the future income of those that do to create a Pareto improvement.

This measure of demand-for-aid can be connected to our social welfare function, by observing that it prioritizes households that have the highest value of $\tau_i P_{it}$, defined as ΔP_{it} , since it represents a discrete change in expected household income. Defining $\frac{\Delta V}{\Delta A_i}$ and $\frac{\Delta V}{\Delta P_{it}}$ as the discrete analogs of the derivative of the value function with respect to aid and permanent income, some algebraic manipulation from equation (2) shows that:

$$\frac{\Delta P_{it}}{\Delta A_i} = \frac{\frac{\Delta V}{\Delta A_i}}{\frac{\Delta V}{\Delta P_{it}}}. \quad (4)$$

When the amount of the aid package is the same for all households, allocating aid to households with the highest ΔP_{it} is equivalent to allocating aid to households with the highest $\frac{\Delta V}{\Delta A_i} / \frac{\Delta V}{\Delta P_{it}}$. This is exactly the result that would occur from maximizing equation (1) with $\alpha_i = (\frac{\Delta V}{\Delta P_{it}})^{-1}$.

This is closely related to previous work on social welfare functions and non-market valuation. It is well known that maximizing a social welfare function weighted by the inverse marginal utility of income is equivalent to maximizing willingness-to-pay (Negishi, 1960). These 'Negishi weights' are often used in non-market valuations to separate distributional questions from questions of efficiency by replicating the market allocation that would occur under complete markets and freezing the distribution of income (Nordhaus and Yang, 1996, Fenichel and Abbott, 2014). In the case of cash aid, however, since cash is fungible with income up to the interest rate, a Negishi weighted social welfare function will just return the market interest rate.

Therefore, instead of weighting by the inverse marginal utility of income, we weight by the inverse marginal utility of *expected future* income. This social welfare function can be seen as freezing the distribution of expected lifetime resources. In other words, the social planner is interested in helping households smooth temporary shocks, but not in redistributing long-run wealth.

Specifying welfare in this way may be regressive in that it gives higher weight to households with high-permanent incomes. On the other hand, wealthy households may be able to satisfy their reconstruction needs by drawing on their savings, or borrowing at a low interest rate. Households with little savings, informal insurance, or access to credit might be willing to pay more to meet their current needs.

Another important consideration is that our framework above uses P_{it} in the time period that the aid is delivered. In some sense this is natural, since it is what would be measured in a post-disaster needs assessment, and it is in fact the variable that we are able to measure in our data. Other possibilities exist, however. We could construct weights using pre-disaster permanent income, which would give a sense of the household's willingness-to-pay to fully insure the disaster. On the other hand, we could use post-aid permanent income, which would be relevant if the aid allowed households to make investments that increased permanent income. In our setting, we find limited evidence of either the disaster or aid having large effects on incomes,³ which is unsurprising since the major effects of both the earthquake and the aid program were on the housing stock rather than productive assets. Section 5.1 performs some sensitivity tests to these other measures of welfare, making assumptions about changes in permanent income with disaster damages and/or the receipt of aid, but given our lack of pre-earthquake income data, we leave a more detailed examination of the quantitative differences between these measures to future work.

One potential objection to specifying welfare in this manner is that households may have high WTP for reasons that have nothing to do with the disaster. A household may be liquidity constrained due to a sequence of bad harvests, or unrelated macroeconomic factors, for example. On the other hand, this can be seen as a positive feature of this SWF, since what is determined to be a 'disaster' may be quite arbitrary (Strömberg, 2007). It's hard to imagine that these arbitrary considerations matter much to the potential recipients of the aid.

To be precise, however, this measure of demand is appropriate for analyzing a policy that aims to improve consumption smoothing, and would be appropriate for a planner who wants to use disaster aid as a form of social-insurance. Thus one way of interpreting these

³See Appendix Table C.I for effects of the earthquake on income, and the regression discontinuity results in Section 4.1 for estimates of the effects of aid on income.

results is as a measure of what fraction of the shocks faced by households in the survey are related to earthquake-induced property damages.

For our main results, we compare various allocations of aid according to both the utilitarian ($\alpha_i = 1$) and WTP for consumption smoothing ($\alpha_i = \frac{\Delta V}{\Delta P_{it}}^{-1}$) criteria. However, we also look at ‘reduced form’ social welfare functions, where the planner only cares about the aggregate effects on rebuilding or on household consumption.

Lastly, it is important to note that, since the data are collected starting from one year after the earthquake, these results are best interpreted as relevant for long-term reconstruction aid, rather than emergency aid. In fact, long-term reconstruction was the primary goal of the major aid program that we analyze. Smaller amounts of aid and in-kind goods were distributed in the immediate aftermath of the earthquake, but most of the total aid budget was disbursed one to three years after the earthquake, which is not uncommon for major disasters.

3.1. *Structural Model of Household Reconstruction*

In order to estimate welfare under various counterfactual targeting policies, we need to put more structure on the household value function. We capture key sources of heterogeneity in a standard dynamic model of incomplete insurance (Deaton, 1991, Kaboski and Townsend, 2011) that incorporates the option to invest in a durable housing stock (Yang, 2009).

Households choose consumption c , and housing investment ι to maximize an infinite discounted stream of utility deriving from consumption and a housing stock, h :

$$U_i = E \left[\sum_{t=0}^{\infty} \beta^t u(c_{it}, h_{it}) \right]. \quad (5)$$

Flow utility takes a standard isoelastic functional form: $u(c_{it}, h_{it}) = \left(\frac{c_{it}^\alpha h_{it}^{1-\alpha}}{1-\gamma} \right)^{1-\gamma}$, that allows for estimation of the importance of housing relative to consumption, as well as risk aversion between periods. h_{it} measures the economic value of the housing stock, which evolves according to:

$$h_{it} = \delta(h_{it-1}) + \iota_{it} \quad (6)$$

where δ is the depreciation factor.⁴ In each period, households receive income Y_{it} , drawn from $\log(Y_{it}) \sim N(\mu_{it}, \sigma^2)$. This allows households to differ in their expected earnings, but they are assumed to have a common variance of log income. Given the income process, household expected income is $\exp(\mu_{it} + \sigma^2/2) := P_{it}$.

In order to smooth consumption, households can borrow and save at interest rate R . Liquid wealth at the beginning of each period is denoted as x_{it} , which evolves according to:

$$x_{it+1} = R(x_{it} - c_{it} - \iota_{it}) + Y_{it+1}. \quad (7)$$

Finally, housing markets are illiquid, and households are subject to a borrowing constraint, B , so households choose c and ι to maximize (5) subject to (6), (7), and:

$$x_{it} - c_{it} - \iota_{it} \geq B \quad (8)$$

$$\iota_{it} \geq 0. \quad (9)$$

The latter constraint ensures that households cannot liquidate their housing stock to fund consumption, which is a realistic feature of rural housing markets in many low-income countries. Because B may be negative, if x_{it} is sufficiently close to B , and $R > 1$, a household could exceed the borrowing constraint, with $x_{it+1} < B$ even if c_{it} and ι_{it} are zero. To avoid this, we follow [Kaboski and Townsend \(2011\)](#) in allowing households to default ($D_{it} \in \{0, 1\}$ is an indicator for default). When $D_{it} = 1$, households receive a minimum consumption and housing bundle:

$$c_{it} = \underline{c}; \iota_{it} = 0; h_{it} = \max[h_{it}, \underline{h}]; x_{it+1} = RB + Y_{it+1} \quad (10)$$

These minimum consumption and housing bundles, as well as the borrowing limit B are assumed to be some fraction of permanent income: $\underline{c} = \bar{c}P_i$, $\underline{h} = \bar{h}P_i$, and $B = \lambda P_i$, where

⁴We assume housing investment is continuous here. This simplification is justifiable in the Nepal context, where many households returned to partially destroyed homes, and reconstruction was an ongoing process. Since households rebuilt their own homes, there were no transaction costs associated with buying and selling homes, a key component of the lumpiness of housing investment in richer countries. However, the default condition described below creates non-convexities in utility with respect to housing investment. In particular, there will be a minimum threshold for investment that generates a mass of households with $\iota = 0$, as we see in the data.

$\bar{c}$, $\bar{h}$, and λ are parameters to be estimated. This option generates non-convexities in household value functions, where households below certain liquidity and housing thresholds will choose to default and consume the minimum bundle. These default zones are visible in Appendix Figure A.1, which plots policy functions for our estimated parameters.

The effects of the earthquake show up through the values of any of the state variables – a destroyed housing stock would result in lower h_{it} , a liquidity shortfall (i.e. as a result of a temporary income shock) would result in lower x_{it} , and reduced expected future earnings would manifest as a lower P_{it} .

Unconditional aid (a_{it}) is modeled as additive to liquidity such that $x_{it+1} = R(x_{it} + a_{it} - c_{it} - \iota_{it}) + Y_{it+1}$. While the aid was supposed to be used for building earthquake resilient housing, this specification allows for a certain degree of fungibility, and for reasonable parameter values, households with low housing stock will use the extra cash for investment. We can also study the value of conditional aid explicitly by forcing households to invest aid in their housing stock.

We normalize all state and control variables by expected income, P_{it} , which reduces the dimensionality of the state space. If $V(\cdot)$ is the value function associated with the maximization of (5), define $\nu(\cdot) = P_{it}^{\gamma-1}V(\cdot)$. This allows us to express x, h, c , and ι as fractions of expected income, which is done below without changing notation. Redefine y_{it} as the log-normal random variable Y_{it} divided by its expected value: P_{it} . Note that $\log(y_{it}) = \log(\frac{Y_{it}}{P_{it}}) \sim N(-\frac{\sigma^2}{2}, \sigma^2)$. For more details see Appendix A.

Combining (5) - (10), the household's problem can be written recursively, with the Bellman equation:

$$P_{it}^{\gamma-1}V(x_{it}, h_{it}|P_{it}) = \nu(x_{it}, h_{it}) = \max_{D_{it}} \left\{ D_{it}\nu_{D=1}(h_{it}) + (1 - D_{it})\nu_{D=0}(x_{it}, h_{it}) \right\} \quad (11)$$

with

$$\begin{aligned} \nu_{D=1}(h_{it}) &= u(\bar{c}, \max[h_{it}, \bar{h}], 0) + \beta E \left[\nu(R\lambda + y_{it+1}, \delta \max[h_{it}, \bar{h}]) \right], \\ \nu_{D=0}(x_{it}, h_{it}) &= \max_{c_{it}, \iota_{it}} u(c_{it}, h_{it}) + \beta E \left[\nu(R(x_{it} - c_{it} - \iota_{it}) + y_{it+1}, \delta h_{it} + \iota_{it+1}) \right] \end{aligned}$$

subject to the normalized versions of constraints (8) and (9).

$\nu_{D=0}(x_{it}, h_{it})$ is the household value function without defaulting, and $\nu_{D=1}(h_{it})$ is the value of defaulting and consuming the minimum consumption and housing bundle.

With this normalization, we can solve for τ , the percentage of expected future income a household would be willing to give up in exchange for aid, in terms of the value function:

$$\tau_i = 1 - \left[\frac{\nu(x_{it}, h_{it})}{\nu(x_{it} + a_{it}, h_{it})} \right]^{\frac{1}{1-\gamma}}. \quad (12)$$

Since τ_i only depends on liquid wealth and housing wealth, it is simple to use equation (3) to study how household WTP for aid changes with different assumptions about how the earthquake or aid changes permanent income. Appendix Figure A.2 plots τ as a function of household state variables. As expected it is decreasing in x_{it} and h_{it} , however, it is also slightly lower in the default zone. This reflects the fact that the marginal utility of liquidity is actually flat in the default zone—households with large debts need a large amount of aid to move to the increasing part of their value functions. This can provide a microfoundation for why some studies of cash transfers have found that households prefer larger lump-sum transfers rather than a stream of payments (Kansikas, Mani and Niehaus, 2025). To analyze the value of conditional aid, we simply add a_{it} to h_{it} instead of x_{it} . As Figure A.2 shows, τ is lower for this type of restricted aid.

The model introduces substantial structure to household decision making. Several assumptions are worth discussing in more detail. First, the model is partial equilibrium, with exogenous income processes and interest rates. While a general equilibrium model would certainly be of interest, given the cost in terms of complexity, a partial equilibrium approach can be justified given that we hold the total amount of aid constant across all counterfactuals. Even if inflows of aid have macro-economic consequences, it is less likely that different allocations of the same amount of aid would have dramatically different implications for credit or input markets.

Secondly, this framework does not allow for externalities resulting from household decisions. In their structural analysis of disaster aid in the aftermath of Hurricane Katrina, Fu and Gregory (2019) find the existence of rebuilding externalities. Agglomeration economies are less important in rural Nepal than in New Orleans, possibly reducing the importance of these externalities. On the other hand, to make their model tractable, Fu and Gregory (2019)'s model features deterministic income, whereas in rural Nepal income

is stochastic and highly variable, justifying a shift in emphasis. Furthermore, since the rental market is virtually non-existent, there are no pecuniary externalities through housing prices, which could be important if a disaster created a housing shortage. Rather than search for market based rental housing, most households lived in temporary shelters close to their original dwellings or moved back into their partially reconstructed homes ([The Asia Foundation, 2016a](#)). A more plausible type of externality of aid in this setting would be increasing liquidity throughout the aid recipient's network, which would make targeting based on observable damages less important.

Finally, the permanent component of income, μ_{it} , is taken as exogenous to choice and state variables. While endogenizing income would be possible with additional parameters (see e.g. [Kaboski and Townsend 2011](#)), this more parsimonious and tractable specification can be justified for two reasons. First, it is consistent with the findings of the post-disaster needs assessment and the survey data discussed above, that emphasized the effects of the earthquake on housing rather than on income or productive assets. We test whether earthquake losses are associated with lower income and can reject even fairly small negative effects (Appendix Table [C.I](#)). Secondly, since the survey data only covers the period following the earthquake, expected income should be thought of as post-earthquake expected income – which is what would be recorded in a post-disaster needs assessment, and what defines our measure of willingness-to-pay. Section [5.1](#) explores some ways of relaxing this assumption, and allowing permanent income to respond to the receipt of aid.

Despite these limitations, the key features of the model are informed by the data and context. Furthermore, when fit to the data, this structure permits an analysis of the relative importance of housing investments, consumption, and savings. The way households make tradeoffs between these choices will allow us to coherently define a notion of a household's demand for aid and analyze the welfare of counterfactual targeting strategies.

3.2. Mapping the Model to Data

In order to estimate individual households' marginal utility and WTP for aid, we map our model of consumption, savings, and reconstruction to the household survey data. The model has three state variables: x_{it}, h_{it}, P_{it} , two continuous choice variables: c_{it}, l_{it} , and an indicator for default: D_{it} . The three state variables summarize the sources of household

heterogeneity. We use the choice variables to define moment conditions to estimate model parameters in the following section.

The WBHRVS data contains detailed consumption information, we construct c_{it} using all categories of non-durables, which is mostly food consumption. The survey includes information on household repair, maintenance, home improvements, and additions, which are taken to be ν_{it} . The survey asks households what they would have to pay to purchase a home like this today. Since this value should include the value of any additions or repairs undertaken during the survey year, we used the lagged value as an estimate of h_{it} . This variable captures earthquake damages well – households that report experiencing earthquake damages have an average housing value 46% lower than households that don't report damages in the first survey wave (before aid was disbursed).

The values of y_{it} are the realizations of household income in each period, which includes wages, agricultural sales (including home food production), business revenues and capital income. P_{it} is estimated as the predicted values of income from a regression of income on a vector of household characteristics, as well as village-year fixed effects (details in Appendix A).

Beginning of period liquidity, x_{it} , includes lagged cash savings, capital gains, income, aid received, new loans taken and informal transfers, including remittances, minus debt. Since the model does not allow for investments in income, we restrict attention to short-term consumption loans (term less than three years and taken within the past three years), and calculate estimated annuities based on each loans' term and interest rate. Similar to [Kaboski and Townsend \(2011\)](#), we define default, D_{it} , as having an outstanding loan balance more than six months after the term of the loan has expired.

Finally, to account for unmodelled variation from the business cycle, and household age structure, size, and education levels, we purge these sources of variation from the data, following the approach in [Kaboski and Townsend \(2011\)](#). Full details are in Appendix A.

Finally, we follow [Kaboski and Townsend \(2011\)](#) by introducing multiplicative measurement error in household expected income. Measurement error is an important concern, and it could affect parameter estimates, since errors that induce fluctuations in income that do not correspond to changes in consumption will make households appear to be more insured than they are in reality. Furthermore, since estimates of household expected incomes are imperfect, the variance of household income shocks in the data may be greater than in re-

ality. For each household, we draw a vector of 100 log normally distributed measurement error shocks with log variance of σ_m . The values of P_{it} are then multiplied by the vector of simulated measurement error shocks. Since all other state and choice are normalized by expected income, this introduces measurement error in those variables as well. We estimate model parameters by matching moments using this simulated data.

4. MODEL ESTIMATION AND VALIDATION

After adding measurement error, the model has ten parameters: $\theta = [\beta, \gamma, \alpha, R, \delta, \bar{c}, \bar{h}, \lambda, \sigma, \sigma_m]$. We estimate these parameters by defining a vector of moment conditions derived from the model and applying the method of simulated moments (MSM) with the optimal weighting matrix (Hansen, 1982).

There are 17 moment conditions in total. The first 9 moments aim to match model predictions of consumption, reconstruction, and default with household behavior in the data, as well as interactions of the deviations in these predictions with household state variables. Another three moments pin down R using data on returns to savings and interest rates payments on loans taken and received. Two moments help to identify σ and σ_m by matching the first and second moments of household income processes to their expectations. Another moment identifies δ using a hedonic style regression on the age of the housing stock. These 15 moments are elaborated in greater detail in Appendix A.

While these moments are consistent with the internal structure of the model, there could be unobserved or unmodelled sources of household heterogeneity that influence parameter estimates. Thus we add two additional moments derived using an exogenous source of variation in access to liquidity. In particular, during the period covered by our survey, aid was targeted at households living in the fourteen most affected districts (see Figure 1). Households close to the borders of these fourteen districts, but on opposite sides of the boundary are expected to have similar levels of earthquake damages, as well as similar geographic, demographic, and economic characteristics. However they differ in their probabilities of receiving aid. This exogenous liquidity shock helps us to pin down policy function responses to a change in liquid wealth. Thus our final two moments take the form:

$$e_{i,16} = \frac{d}{da_{it}} \log c^*(x_{it}, h_{it}, P_{it}) - E\left(\frac{d \log c_{it}}{da_{it}}\right) \quad | \quad a_i > 0 \quad (13)$$

$$e_{i,17} = \frac{d}{da_{it}} \log \iota^*(x_{it}, h_{it}, P_{it}) - E\left(\frac{d \log \iota_{it}}{da_{it}}\right) \quad | \quad a_i > 0.$$

The first term on the right side of each equation represents the change in the log of the optimal policy function response to a change in liquidity. It is calculated by taking the difference between the log of model predicted consumption and housing investment with observed liquidity, minus any aid received, and the log of the predictions with observed liquidity, plus 300,000 NPR, corresponding to the size of the aid package. For $e_{i,17}$, we take the log of $\iota + 1$ to deal with zeros. The following sections describe how we estimate $E(\frac{d \log c_{it}}{da_{it}})$ and $E(\frac{d \log \iota_{it}}{da_{it}})$, the causal effects of aid on consumption and housing investment. Since we can only estimate these quantities in our data for households that received aid, which was conditional on housing damages,⁵ we only estimate these moments for households that received aid ($a_i > 0$).

4.1. Regression Discontinuity Estimates of the Effects of Aid

Receipt of aid is endogenous to observed and unobserved earthquake damages, which are likely to be correlated with outcomes of interest, including consumption, income, savings, and housing investment. In addition, the earthquake occurred in a distinctive region of the country – disproportionately affecting the mountainous districts surrounding Kathmandu. To estimate the causal effects of aid, we rely on an administrative feature of the reconstruction aid program – aid was prioritized for the fourteen most affected districts. In 2018, 60% of eligible households in the most affected districts (red and orange in Figure 1) had received at least 2 tranches of aid, whereas, only 15% of eligible households in the moderately affected districts (yellow, green, and blue in Figure 1) had received the same (Housing Recovery and Reconstruction Platform, 2018).

These features of the response suggest the use of a spatial regression discontinuity (RD) approach to compare outcomes between households close to the borders of the designated districts. Thus we estimate the following regression:

$$Z_{it} = \beta_1 \widehat{\text{Aid}}_{it} + \beta_2 S_i + \beta_3 S_i \mathbf{1}_{\{S_i > 0\}} + e_{it}. \quad (14)$$

⁵In fact, section 5 highlights how the estimated causal effects of aid change according to different targeting strategies.

Z_{it} is an outcome variable for household i at time t . Aid_{it} is a dummy variable for whether a household has received at least their first tranche of aid by time period t , and it is instrumented with a dummy variable for whether a household is on the ‘right’ side of the border ($\mathbf{1}_{\{S_i > 0\}}$), making this a fuzzy RD specification. β_1 thus gives us $E(\frac{d \log Z_{it}}{da_{it}})$, which, when Z is consumption and housing investment in turn, gives us the estimates we need to construct the final two moment conditions.

The running variable S_i is distance from each household to the western border of the 14 most affected districts.⁶ The specification allows the slope of the running variable to differ on either side of the border. [Cattaneo, Idrobo and Titiunik \(2020\)](#) suggest using distance to a specific point on the border for two dimensional regression discontinuity applications rather than distance to the entire border, so we also show a specification that uses distance to the point on the border that is closest to a village.

In some specifications we include a vector of control variables, which include self-reported earthquake damages, age and education of the household head, a high-caste dummy, and the travel time to the nearest health clinic.

Since the RD analysis restricts the regression to a subset of ‘border’ households, one of the crucial parameters for the RD analysis is determining the bandwidth within which a household is included in the regression. A wider bandwidth typically allows for more precision, at the cost of biased estimates if there is curvature in the slope of the running variable near the border point. We use the methods from [Calonico et al. \(2017\)](#) and [Calonico, Cattaneo and Farrell \(2020\)](#) to calculate the optimal bandwidths for point-estimation and inference, which vary for each dependent variable, typically falling between 25 and 60 km in the main specifications. We test robustness to alternate bandwidths in Appendix B.2. The blue dots in Figure 1 indicate villages that fall within the bandwidth for inference for the first stage regression – 60 km. All regressions use kernel times survey weights. The kernel weights give more weight to households closer to the border, allowing us to control for dis-

⁶We show robustness to using the entire border in Appendix B. The western border is used as it is close to the epicenter of the earthquake and features comparable geography and earthquake damages across the border. The southern border of the 14 districts is the geographic boundary with the Terai (plains region of Nepal), and earthquake damages were much lower in these districts, as well as along the eastern border.

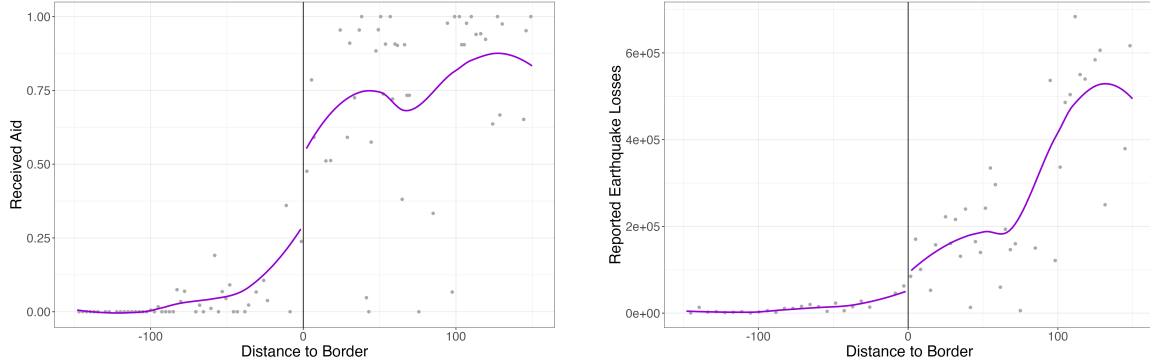


FIGURE 2.—Binned averages of fraction of households receiving aid and self-reported earthquake damages as a function of distance to the border with Loess smoothing.

tance to the border non-parametrically.⁷ We use heteroskedasticity-robust standard errors following the recommendations of [Kolesár and Rothe \(2018\)](#) in settings where the running variable is discrete.

Figure 2 shows the identification strategy graphically. The left panel shows the binned averages of the probability of receiving aid as a function of distance to the border point. There is a clear jump in the probability of receiving aid at the border. The right panel shows self-reported earthquake damages plotted in the same way – there is no comparable discontinuity. The results of the first stage regression show that households just inside the border of the most affected districts were 23% more likely to receive aid, and received nearly 40,000 NPR more aid on average in the primary specification (Appendix Table B.VI). Alternative specifications give mostly similar results, including different choices of bandwidth, kernel, distance to alternate border points, inclusion of less affected districts, inclusion of control variables, and a ‘donut hole’ specification that excludes villages within 5 km of the border.

4.1.1. Identification Assumptions

For β_1 to accurately identify the causal effects of aid, it must be the case that the receipt of aid is the only thing that changes discontinuously at the border. Other political, institutional, and geographic factors must be a continuous function of geography, and in particular there

⁷We use triangular weights in the baseline specification and test sensitivity to alternative kernels in Appendix B.

cannot be sorting across the border – a potent concern in spatial RD settings as discussed by [Keele and Titiunik \(2015\)](#).

These assumptions hold in this setting for two reasons. First, pre-quake political institutions in Nepal were weak as discussed in Section 2. The district borders were mostly set in the 1960s during the Panchayat Regime – a system of governance in which district representatives were elected to serve on a partyless National Assembly that was powerless to do anything other than rubber-stamp the King’s agenda. For these reasons, there shouldn’t be large differences in policy across district boundaries. Furthermore, district borders do not demarcate boundaries between ethnic groups, giving us no reason to believe that informal institutions change discretely at the border either ([Gurung, Gurung and Chidi, 2006](#)).

Regarding sorting, land and housing markets in rural Nepal are not very liquid. While rural to urban and international migration are important livelihood strategies, rural-to-rural migration is rare. If entire households migrated away from regions that did not receive aid, this could be problematic. It is much more common for one or two individuals within a household to migrate and send back remittances, however.⁸

As suggested by [Imbens and Lemieux \(2008\)](#), we can gain confidence in the underlying RD assumptions by conducting placebo tests on household demographic, political, and geographic variables. To do so we run versions of equation (14), replacing Z with placebo variables that should not change with the receipt of aid. For this exercise, we compare households baseline characteristics using data from the first wave of the survey, immediately after the earthquake, but before aid had been disbursed.

Most household demographic variables appear smooth at the border, although some are noisy (Appendix Figures B.4). Table B.VII formally tests for discontinuities, and finds no significant differences in self-reported damages, caste, age, the number of household members, the probability that a household member has completed 5 or 10 years of schooling, or the fraction of households that have always lived in the same house or same district on either side of the border, supporting the idea that household demographics are comparable. While there is no discontinuity at the border, there is a decreasing share of households that

⁸A McCrary test for discontinuities in the density of households on either side of the border is not informative in this setting because the sample frame for the survey is based on the 2010 census, but sampling occurred after the earthquake in 2015 ([McCrary, 2008](#)) We plot the density of households as a function of distance to the border anyway and do not observe any obvious discontinuities (see Appendix B).

have always lived in the same district as distance from the border increases in the most affected districts. This is not likely to be due to earthquake damages or aid causing differential in-migration, but rather these districts are closer to Kathmandu and thus have seen more in-migration over time. Indeed, we show below that, if anything, migration decreases with receipt of aid. We also find no differences in the receipt of NGO aid or non-earthquake government transfers, supporting the hypothesis of no policy changes at the border.

We test for differences in prices of common food items, as well as for travel times to the nearest bank, school, market, and healthcare clinic as a proxy for market integration and public good provision, and we find no differences. Differences in elevation across the border are not significant, but there is a difference in slope that is significant at 5%. This significance does not survive using [Calonico, Cattaneo and Farrell \(2020\)](#) robust confidence intervals, and correcting p-values for multiple hypothesis testing, however.⁹

4.1.2. *The Causal Effects of Disaster Aid*

Turning attention to outcomes of interest, the main specification in the first row of Table III shows large effects of aid on log consumption and home investment, and large negative effects on remittances and migration.¹⁰ Effects on savings, loans, investments, and income are not distinguishable from zero. Figure 3 shows the discontinuities for these variables in the data, which are clearly visible for housing investment and remittances. Similar plots for additional variables, including the placebos, appear in Appendix B.2.

We show robustness to alternative choices of kernel, bandwidth, dependent variable functional form, and the inclusion of demographic and geographic control variables in Appendix B.2. Main results on consumption, investment and remittances are robust to these variations. In some specifications we find a positive and significant effect effect on new loans taken, and/or income. We test robustness of the results to using different definitions of the running variable, such as including the eastern and southern border segments, or using

⁹Using distance to the single point, rather than distance to the full border shows significant differences in self-reported damages, NGO transfers, and some of the travel time and geographic variables, however none of these differences remain significantly different from zero using [Calonico, Cattaneo and Farrell \(2020\)](#) robust confidence intervals and correction for multiple hypothesis testing.

¹⁰After using [Calonico, Cattaneo and Farrell \(2020\)](#) confidence intervals and correction for multiple hypothesis testing, the effects on consumption and home investment remain significant at 5%, though the p-values for the effects on remittances and migration are above 10%.

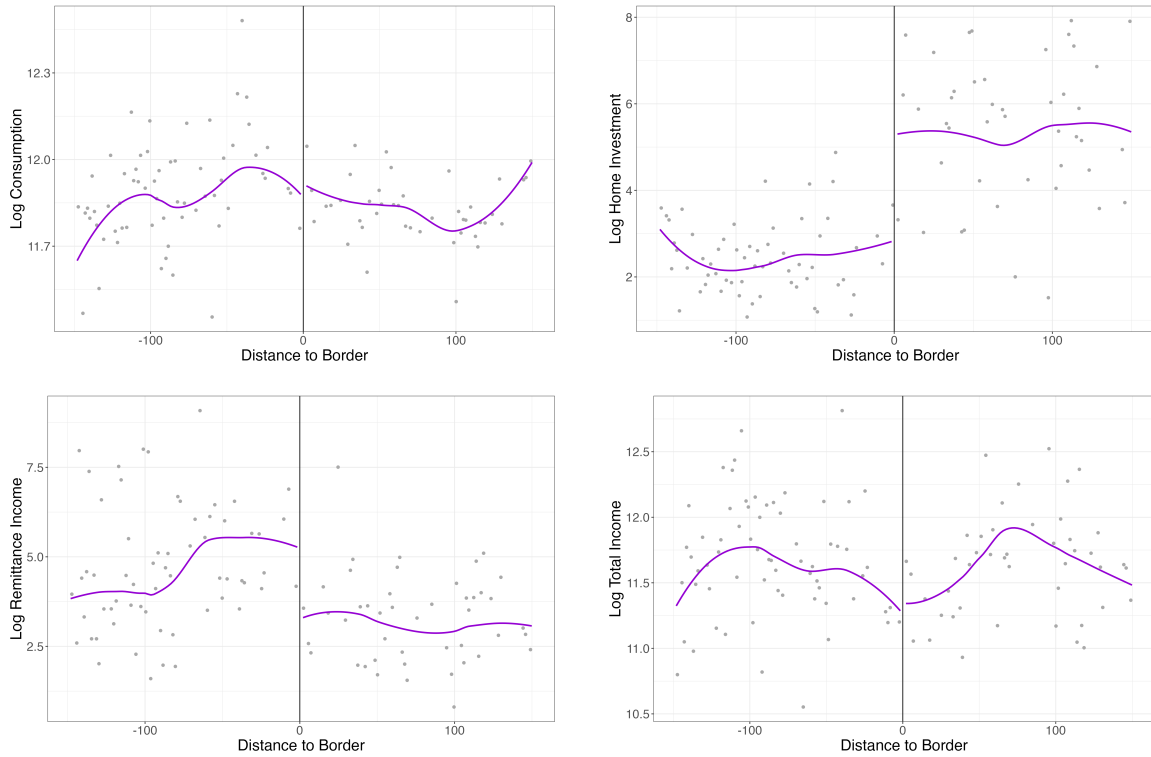


FIGURE 3.—RD Plots: Binned averages of outcome variables in logs as a function of distance to the border with Loess smoothing. Positive values indicate villages inside the districts that were prioritized for aid. Added 1 to remittances and home investment to account for zeros.

distance to a specific point on the border. This leads to noisier estimates for all variables, especially when the southern border is included, which includes districts in the plains region of Nepal that have a very different geography and minimal earthquake damages (see Figure 1).

These estimates are qualitatively consistent with the structural model's predictions. Households increase consumption and investment, showing that aid is somewhat fungible, even though it was supposed to be used for housing reconstruction. Aid also substituted for other sources of liquidity, in particular remittances, which are an important source of informal insurance in Nepal, as discussed in Section 2. Given the estimates on new loans taken, we have some weak evidence that aid may have induced additional borrowing, which would be consistent with additional liquidity relaxing a borrowing constraint. These results support the model being a reasonable approximation of household decision making in this context, and thus, once properly calibrated, can be informative about household values

Regression Discontinuity Estimates: Distance to Western Border								
	consumption (1)	home investment (2)	remittance income (3)	cash savings (4)	new loans taken (5)	investments (6)	total income (7)	migration (8)
$\widehat{Aid}$	0.66** (0.32)	9.79*** (3.77)	-6.47** (3.20)	0.90 (2.54)	5.69 (3.57)	1.25 (2.56)	0.93 (0.61)	-0.76** (0.37)
N	1042	1435	695	1071	916	1310	476	1093
Bandwidth	33.89	42.28	25.33	34.14	31.34	38.83	22.05	34.92
Wards	66	66	35	46	40	77	35	45
Regression Discontinuity Estimates: Distance to Single Point								
	consumption (1)	home investment (2)	remittance income (3)	cash savings (4)	new loans taken (5)	investments (6)	total income (7)	migration (8)
$\widehat{Aid}$	0.34* (0.20)	4.15* (2.13)	-5.13* (2.97)	0.85 (1.88)	4.13* (2.21)	0.37 (1.82)	1.26* (0.66)	-0.38 (0.26)
N	432	1873	1047	1550	1793	1658	1220	1224
Bandwidth	29.14	59.68	41.67	50.47	56.39	53.51	43.02	42.74
Wards	34	85	55	76	79	85	57	57
Model Predictions: Response to 300k Aid								
	food consumption (1)	home investment (2)	borrowing (3)					
Aid	0.20 (0.24)	1.73 (2.58)	-5.89 (9.50)					

TABLE III

POLICY FUNCTION ESTIMATES

Top panel shows regression discontinuity estimates using local linear regressions with triangular kernel times survey weights, heteroskedasticity-robust standard errors, and optimal bandwidths. Slope of running variable allowed to differ on either side of cutoff. Distance > 0 used as instrument for aid. All dependent variables in logs, with 1 added to home investment, remittances, cash savings, loans, investments, and migration to account for zeros. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels respectively. Bottom panel shows calibrated model predicted effects on choice variables for households that actually received aid.

Population estimated standard deviations in parentheses.

for aid. In calibrating the model, we take our estimates from the first two columns of Table III to construct moment conditions e_{16} and e_{17} , setting $E(\frac{d \log c_{it}}{da_{it}} | a_i > 0) = 0.66$, and $E(\frac{d \log l_{it}}{da_{it}} | a_i > 0) = 9.79$.

4.2. Model Validation

The parameter values that minimize the MSM objective function are shown in Table IV. In comparison to Kaboski and Townsend (2011), we find a notably high coefficient of risk aversion, and a low discount factor, indicating rather extreme risk aversion and present bias. This difference arises for a few reasons. First, Kaboski and Townsend (2011) did not observe cash holdings, so we have a more comprehensive measure of liquid wealth. Holding more wealth makes households look more risk averse. Secondly, the structure of the

economy in Nepal is different, with much higher reliance on remittances. Thus liquidity-to-permanent income ratios are much higher in this context. This also explains why we find a borrowing constraint (λ) that is relatively high as a fraction of permanent income. Finally, previous work has suggested that exposure to natural disasters may increase risk-aversion (Cameron and Shah, 2015). It is plausible that this mechanism is also at play in this setting. We use these values for our primary counterfactuals, but we show robustness of our results to using risk-aversion and discount-factor parameters from Kaboski and Townsend (2011) in Section 5.1.

In terms of the other parameters, interest rates are very low, which is common in a low-income context, as few households earn interest on their cash holdings. Depreciation rates and the Cobb-Douglas share on consumption are within the range of normal values reported in the literature, and the minimum consumption bundle ($\bar{c}$) is comparable to what Kaboski and Townsend (2011) estimate. The minimum housing bundle associated with default ($\bar{h}$) is a new parameter, so we lack relevant sources of comparison. For a household at the average income level, this indicates a value of housing stock in default at about \$1,800. Median and mean home values in the survey are 2.9 and 6.8 times larger respectively. This estimate could be consistent with reports suggesting many earthquake-affected households moved in with friends and family, or moved into partially reconstructed houses, since it is about 4 times the cost of an emergency tent (ICRC, 2025). Since results about targeting based on housing damage are likely to be sensitive to household's outside options for housing, we test sensitivity to a lower value of this parameter in Section 5.1.

Using the estimated model, we can calculate each household's expected marginal utility and WTP for aid,¹¹ which will allow us to compare aggregate welfare from different targeting strategies. But first we conduct a series of validation checks to ensure that the model is capturing something real in the way households value aid.

¹¹We calculate expectations by integrating over measurement error (s_{it}): $E[WTP_{it}] = \int WTP(x_{it}, h_{it}, P_{it}, s_{it}) ds_{it}$. We use the same approach for marginal utility and the expected consumption and investment responses.

Variable	Definition	Source
c_{it}	Consumption	Value of last week's food consumption in survey multiplied by 52, plus annual spending on utilities, transportation, clothing, and entertainment.
ι_{it}	Housing Investment	Spending on home repairs, improvements, maintenance, furniture, and large appliances over the last 12 months.
D_{it}	Default	1 if a household has unpaid debts more than 6 months past the term of the loan, otherwise 0.
Y_{it}	Realized Income	Income from wages, agricultural sales, food home production, self-employment, land, housing, and equipment rentals, and pensions and other government programs.
P_{it}	Expected Income	Estimated from household characteristics, see Appendix A.
x_{it}	Liquid Wealth	Lagged cash savings, capital gains, income, aid received, new loans taken, informal transfers, and remittances, minus debt.
h_{it}	Housing Wealth	Lagged estimate of self-reported value from survey. Age of housing stock also used to estimate depreciation.

Parameter	Estimate (SE)	Interpretation
γ	4.88 (0.06)	Coefficient of Risk Aversion
β	0.89 (0.02)	Discount Factor
α	0.78 (0.05)	Cobb-Douglas Share on Consumption
δ	0.95 (0.01)	Depreciation Factor for Housing
R	1.002 (0.001)	Interest Rate (+1)
$\bar{c}$	0.57 (0.16)	Minimum Consumption as Fraction of Expected Income
$\bar{h}$	0.65 (0.19)	Minimum Housing as Fraction of Expected Income
λ	0.57 (0.09)	Borrowing Constraint as Fraction of Expected Income
σ	1.03 (0.01)	Standard Deviation of Log Income
σ_m	0.12 (0.10)	Standard Deviation of Log Measurement Error

TABLE IV

MODEL VARIABLES AND ESTIMATED PARAMETERS

Structural model parameter estimates from method of simulated moments using optimal weight matrix. See text and Appendix A for details. Parameter standard errors clustered by household.

In terms of targeted moments, the third row of Table III compares the regression discontinuity estimates to the model predictions by calculating expected counterfactual consumption, investment, and borrowing responses to aid using the optimal policy and investment functions from Section 4. We subtract any aid received from each household’s liquid wealth and calculate predicted consumption, investment, and borrowing. Then we add back 300,000 NPR, and calculate consumption, investment, and borrowing again.

The model predicts smaller effects on consumption and housing investment than we observe in the RD, although the confidence intervals from the RD estimates are wide. The model’s predictions are closer to the effects estimated using the single point cutoff design, suggesting that there could be treatment effect heterogeneity. This is what the model predicts, as shown by the large standard deviations in the effects of aid on consumption and investment in Table III. Although it was not used as a targeted moment, the model’s predicted effect on borrowing is well aligned with the estimated effects on remittances from the RD.

Since our objective is to study how different targeting policies correlate with demand for liquidity to smooth consumption, we validate our estimates of WTP, by looking at whether the values correlate with household characteristics and behaviors that relate to consumption smoothing. Table V shows regressions of estimated WTP on household characteristics in panel A and strategies for coping with shocks in panel B.

In panel A, we see that Newari households have the lowest estimated WTP for aid, and Dalit and other castes have higher WTP. Land-rich households also have lower WTP for aid – they are willing to give up 3.3% less of their expected future income in exchange for liquidity now. These findings are consistent with Table I, which shows that Newari and land-rich households are better able to smooth consumption. The coefficient on female headed households is not statistically different from zero—while many of these households have access to remittances, this may be offset by households that may have sent a recent migrant in response to the earthquake. The final column shows that earthquake-affected households have 1.8% higher WTP for aid, indicating that our measure is capturing the distress experienced by these households.

In panel B, we perform further validation checks by seeing whether our measure of WTP is correlated with costly self-reported household strategies for coping with shocks. Column 1 uses as the independent variable the maximum interest rate on any outstanding household

loan. Households that are paying higher interest rates have higher WTP, supporting the interpretation of WTP as demand for liquidity. Column 2 and 3 show that WTP is higher among households that reported cutting on food consumption or having a child under the age of 18 working. In particular, households that report having a child working are willing to give up 2.4% more of their future income in order to access aid now. The final column shows that households receiving remittances have 3.3% lower willingness to pay for aid. Although these regressions are descriptive, the findings support the interpretation of WTP as the demand for liquidity to smooth consumption.

5. COUNTERFACTUAL REALLOCATIONS

We analyze a variety of targeting policies, as well as the observed allocation, using our measures of WTP to assess the aggregate surplus created by different aid allocations. We normalize welfare relative to a baseline “random allocation” scenario, which sets benefits equal to the population average WTP.

As discussed above, household WTP is increasing in expected future income. Thus, an allocation could result in low welfare gains if it prioritizes low-income households. Therefore, we also compare targeting strategies using a utilitarian social welfare function. This criterion prioritizes households with a high marginal utility for liquidity, which tend to be the poorest households. Furthermore, we show the aggregate effects of each allocation on consumption and housing reconstruction, which would be relevant for a planner interested in prioritizing food security or externalities associated with housing reconstruction respectively. Full results for all welfare measures and targeting schemes are in Table VI.

All counterfactuals are analyzed based on the second survey wave, which is when most aid-eligible households received the first tranche of reconstruction aid. This will somewhat overstate the value of the actual allocation of aid, since some households didn’t get the aid until later, and those that did received aid in several tranches over multiple years. Our approach analyzes the value of receiving all the aid at once. We calculate household liquidity without including any aid received in that period. Since we use lagged measures of the housing stock, that state variable reflects the state of the housing stock in the first survey wave, right after the earthquake, and is thus uncontaminated by the effects of aid. We restrict the analysis to the earthquake affected districts (including moderately and slightly affected districts) to set a reasonable restriction on the eligible population. In these districts,

A. Household Characteristics				
	WTP as Fraction of Income in Percentage Points			
	(1)	(2)	(3)	(4)
(Intercept)	17.76*** (0.50)	18.43*** (0.34)	20.21*** (0.41)	17.56*** (0.45)
Dalit	2.57*** (0.98)			
Newar	−1.73 (1.41)			
Other	1.27* (0.65)			
Female Head		0.68 (0.69)		
Land > Median			−3.31*** (0.58)	
Quake Affected				1.77** (0.59)
Num.Obs.	2085	2085	2085	2085
R2	0.006	0.000	0.015	0.004

B. Household Coping Strategies				
	WTP as Fraction of Income in Percentage Points			
	(1)	(2)	(3)	(4)
(Intercept)	17.54*** (0.41)	18.53*** (0.30)	18.31*** (0.31)	19.89*** (0.37)
Max Interest Rate	0.08*** (0.02)			
Cut Food Consumption		2.41 (1.90)		
Children Worked			2.39** (0.91)	
Received Remittances				−3.34*** (0.60)
Num.Obs.	2085	2085	2085	2085
R2	0.006	0.001	0.003	0.015

TABLE V

REGRESSIONS OF WTP FOR AID AS A PERCENT OF FUTURE INCOME (IN PERCENTAGE POINTS) ON HOUSEHOLD CHARACTERISTICS (PANEL A) AND SELF-REPORTED COPING STRATEGIES (PANEL B). Household characteristics are defined as in Table I. Coping strategies are the maximum interest rate paid on an outstanding household loan, and dummies for whether a household reports cutting food consumption, having a child under the age of 18 engaged in agricultural or non-agricultural labor, and receiving remittances. Sample is second wave of WBHRVS, households in earthquake affected districts, the same as what is used for counterfactuals. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels respectively.

58% of households reported experiencing earthquake damages, and 38% of households received aid.

In all counterfactuals, except for the universal allocation, we hold the fraction of households receiving aid constant. In the universal allocation, we hold the budget constant, by making the amount of aid to each household $0.38 \times 300,000 = 114,000$ NPR. Survey weights are used to make the estimates representative. 95% confidence intervals for each targeting strategy are bootstrapped using 1,000 replicates, and reflect sampling error, but not specification error from the model or parameter estimates.

We find that the average household would give up 18.6% of future income in order to receive 300,000 NPR in reconstruction aid, reflecting a WTP of 431,523 NPR in present value – 44% higher than the nominal value of the aid.¹² There is substantial heterogeneity, however, with a standard deviation of nearly 310,000 NPR. If we make aid conditional, such that the household must invest all of it in their housing stock, WTP is on average 49% lower than for unconditional aid. If a planner was able to perfectly identify the households with the highest WTP for aid, this optimal allocation of aid would improve WTP by 72% relative to a random allocation. Since this allocation may be difficult to achieve, we test alternative strategies based on simple proxies that are easier to observe.

We find that the WTP of households that actually received aid was 9% better than the random allocation. Since aid eligibility was conditional on the engineers' assessments that the house was completely destroyed, this assessment should be considered to take into account information about the relative damages of the structure, but we also analyze a scenario based on targeting households with the largest absolute damages based on self-reports. We find that the welfare generated from this approach is very close to the observed allocation, with an identical point estimate. Neither of these allocations perform well by the utilitarian criterion, however, yielding substantially less welfare than the random allocation. They do increase housing reconstruction by 12-13% more than the random allocation, however.

Columns 3 and 4 show the benefits of targeting the households with the lowest consumption and liquid wealth, respectively. These households are the typical targets of anti-poverty cash transfer programs, and unsurprisingly, targeting them performs well according to the utilitarian criterion. In fact, targeting the lowest liquidity households approaches the wel-

¹²Present value calculation uses the estimated discount factor of 0.89.

Welfare Measure	Targeting Scenario						
	Actual (1)	Damages (2)	Consumption (3)	Liquidity (4)	Village Based (5)	Universal (6)	Optimal (7)
WTP	1.09 (1.04, 1.12)	1.09 (1.05, 1.12)	0.99 (0.95, 1.03)	0.97 (0.91, 0.99)	1.12 (1.07, 1.15)	1.11 (1.09, 1.12)	1.72 (1.64, 1.74)
Utilitarian	0.44 (0.11, 0.68)	0.50 (0.11, 0.78)	1.34 (0.69, 1.87)	2.45 (2.31, 2.62)	0.24 (0.04, 0.36)	0.88 (0.51, 1.24)	2.64 (2.54, 2.82)
Consumption Increase	1.02 (0.95, 1.10)	1.00 (0.92, 1.06)	0.95 (0.88, 1.02)	0.41 (0.28, 0.45)	1.09 (1.01, 1.15)	1.00 (0.95, 1.06)	2.28 (2.18, 2.42)
Housing Increase	1.12 (1.01, 1.19)	1.13 (1.01, 1.19)	1.08 (0.98, 1.16)	1.20 (1.13, 1.31)	1.12 (1.02, 1.20)	1.06 (0.99, 1.14)	2.36 (2.25, 2.48)

TABLE VI

COUNTERFACTUAL TARGETING SCENARIOS

Benefits relative to a random allocation for 300,000 NPR aid allocated to 38% of the population according to various targeting strategies – except universal scenario which uses 0.38*300,000 NPR allocated to the entire population. Bootstrapped 95% confidence intervals in parentheses using survey weights. Utilitarian refers to an ‘equal-weighted’ utilitarian social welfare function. Consumption and liquidity allocations based on households with the lowest values of those variables. Damages based on households with the highest self-reported earthquake damages. Village based gives aid to all households in villages with the highest peak-ground acceleration from USGS.

fare gains of the optimal allocation by the utilitarian criterion. According to the WTP criterion, the welfare generated by these allocations is slightly less than the random allocation, however, showing a clear tradeoff between the two welfare measures. Neither of these allocations tends to increase consumption by much, since they tend to target households with low incomes, and thus low-levels of steady-state consumption. Liquidity targeting generates a 20% increase in housing reconstruction, however, which is more than targeting based on damages. This shows that it is not just the earthquake-affected households that would like to invest in their housing stock.

In column 5, we study a targeting policy that gives aid to all households in villages that experience the most intense shaking from the earthquake, whether or not their house was destroyed. Interestingly, this policy increases welfare by 12%, slightly more than the actual allocation. The universal allocation (column 6) does similarly well, increasing welfare by 11%. These estimates do not account for savings associated with targeting and administration of the allocation.¹³ Neither of these allocations do well by the utilitarian

¹³Hanna and Olken (2018) argues the administration costs of targeting are typically low – on the order of 1% of the program budget, so this is not a large source of gains. Given our estimates of the discount factor in Table IV,

	Average WTP	N Households
Earthquake	1.05	1,251
Drought and Other Environmental Shocks	1.02	53
Livestock Disease or Crop Failure	1.02	123
Illness or Injury in Household	0.94	567
Riots	0.88	368

TABLE VII

WILLINGNESS TO PAY FOR AID BY HOUSEHOLDS EXPERIENCING DIFFERENT TYPES OF SHOCKS (AS A FRACTION OF AVERAGE WTP).

welfare measure, although the universal allocation out-performs the actual and damage-based allocations. Village-based targeting has the largest impact on food consumption, and a comparable effect on housing reconstruction as the damage-based allocations.

The relative success of the universal approach reflects the concavity of household value functions over liquidity, which generates diminishing WTP for aid. Giving all households a little bit of aid moves them up the steep section of their value functions. Intuitively, both the universal and the random allocation ‘waste’ some aid by giving it to households that don’t need it. But the universal allocation makes sure that all households can cover some basic needs, while a random allocation may miss some very needy households, while giving larger amounts to less needy households.

One possible explanation for the relative success of the shake intensity and the universal targeting policies, neither of which conditions aid on housing damages, is that, as previously stated, our measure of WTP is not earthquake specific, and thus other types of shocks could influence household WTP. Table VII shows the average WTP for households that experienced different types of shocks, and the number of households that experienced each type of shock. The results show that on average, households experiencing the earthquake had higher WTP than households reporting other types of shocks. This suggests that our results can be explained more due to heterogeneity in access to liquidity rather than the importance of other types of shocks.

These results show a strong tradeoff between policies that target households with high marginal utility for cash versus policies that target based on WTP for smoothing the shock.

getting aid distributed more quickly could be a more important source of welfare improvements, however, which is not accounted for by the estimates in this section.

It is interesting to note, however, that the actual and damage-based allocations are not best by either metric, and in fact, they are strictly dominated by the universal allocation with respect to both metrics, which also has the added benefit of being easy and fast to implement. For policymakers looking to increase aggregate food consumption, village-based targeting is preferable to damage-based targeting, and it is virtually identical in terms of generating housing reconstruction.

5.1. Sensitivity Analyses

5.1.1. Alternate Measures of Welfare

As discussed in Section 3, an alternative to measuring WTP based on permanent income estimated in the period when aid is delivered, is to measure WTP based on either pre-earthquake permanent income, or post-aid permanent income. This is simple to compute, since we have already estimated τ_i , we simply need to adjust equation 3 with the relevant estimate of P_{it} .

Using pre-earthquake income could be important if exposure to the disaster has a durable negative effect on incomes. We do not observe pre-disaster incomes, but in Appendix C, we regress household damages on income, using shake intensity as an instrument, and we find that, if anything, incomes are higher in earthquake exposed households, even in the first wave of the survey, before households had received aid. Still, as a best-case scenario for damage-based targeting, we test robustness to the assumption that households with earthquake-related housing damages had 10% higher pre-quake permanent incomes.

In this case, both the actual and the damage-based targeting scenarios perform better, increasing welfare by 12% and 13% relative to the random allocation respectively. Village-based targeting does even better, however, increasing welfare by 14% relative to the random allocation. The universal scenario is unchanged, with an 11% increase. See Appendix D for full results.

We also consider using post-aid permanent income. While the regression discontinuity analysis doesn't find robust effects on income, the point estimates are large and noisy. Applying a 0.93 log point increase in incomes to all households doesn't change the relative welfare of targeting schemes, but it does change the magnitude of estimated WTP to 1.1 million NPR, now more than 3.6x nominal value of the aid. Heterogeneity in income effects

could change the relative rankings of various targeting schemes, but it is probably less likely that the households that need to spend the aid rebuilding their homes will also have access to more profitable investment opportunities.

5.1.2. *Alternate Parameters*

Since our estimates of the coefficient of risk aversion ($\gamma = 4.88$) and discount factor (0.89) indicate quite strong present bias and risk aversion relative to previous estimates in the literature, we test the sensitivity of our results to these parameters, instead using the [Kaboski and Townsend \(2011\)](#) values of $\gamma = 1.2$ and $\beta = 0.926$. A high coefficient of risk aversion generates a steep utility function at low levels of wealth and rapidly diminishing WTP for aid as wealth increases. This makes universal targeting a relatively more attractive strategy, since a smaller cash distribution, given to everyone, can ensure that all households cross this steep section of the utility function. Table D.I in Appendix D confirms this intuition. Using the lower coefficient of risk aversion decreases the relative benefits of universal aid dramatically to a 1% gain relative to random, and increases the benefits of the actual and damage-based allocations to 10% each. The village-based strategy still outperforms damage-based targeting, however, with a 14% welfare gain.

Given the lack of previous estimates in the literature of the minimum housing stock parameter $\bar{h}$, and our relatively imprecise estimate, we also study sensitivity to this parameter, setting it to 0.326, half of our previous estimate, and just outside the lower end of our 95% confidence interval. This increases the relative benefits of the actual, damage-based, and shake-intensity allocations by 1% each. This is unsurprising, as reducing the value of the outside option for housing increases the value of targeting households with housing damages. The village-based and universal allocations still perform better, however. Intuitively, the lower value of $\bar{h}$ makes the value function steeper at lower levels of housing and liquid-wealth, in some ways mimicking the effects of a higher risk-aversion parameter. Full results are in Appendix D.

5.1.3. *Different Distributions of State Variables*

Lastly, we perform a simple exercise to study how our results might depend on changes in inequality or average levels of wealth. We adjust all three state variables $q_{it} \in (x_{it}, h_{it}, P_{it})$ according to $\tilde{q}_{it} = w(q_{it} - \bar{q}) + gq_{it}$, where w controls the spread of the distribution (with

mean $\bar{q}$ before the transformation) and g determines the mean of the new distribution. For each distribution, we calculate the ratio of aggregate welfare from the actual targeting policy followed after the Nepal earthquake to the welfare under the universal allocation. Full results are in Appendix D.

While this is highly stylized and simplistic, in that it holds constant the allocation of aid and the parameters of the value function, it can still give a sense of how the value of targeting is sensitive to the distribution of household wealth. Intuitively, an increase in inequality increases the heterogeneity in values for aid, and thus, could make targeting more valuable. This is indeed what we observe in the simulations. Holding g constant at 1, the value of targeting relative to universal aid goes from 11.4% lower when inequality is halved, to 3.0% higher when inequality is doubled. This pattern of universal aid outperforming targeting at low levels of inequality, and underperforming at high levels of inequality holds consistently across different values of g .

The effect of shifts in the average state variables is more subtle and non-monotonic. Keeping $w = 1$, universal targeting is 1% better when $g = 1$, 0.9% worse when $g = 3$, and 5.4% better when $g = 4$. This pattern can be explained by nonlinearities in the value function - at highly negative levels of liquidity, aid has no value, since households default regardless. Aid has the highest value for households close to the default zone or just above it, and then it decreases again as housing and liquid wealth increase. The results thus indicate that targeting has highest value at high levels of inequality and intermediate levels of average wealth.

6. CONCLUSION

Natural disasters may present households with shocks that existing institutional arrangements are incapable of smoothing. This is especially likely when the disaster is outside of recent lived experience, and when the disaster occurs in a low-income country, where credit constraints are more likely to bind and informal risk sharing networks can be overwhelmed by a large covariate shock. In these circumstances, aid can facilitate reconstruction and consumption smoothing by providing liquidity to ease these constraints. This paper explores the value of targeting aid in the context of the 2015 Nepal earthquake, taking an approach grounded in welfare theory.

Targeting aid could be important if households differ in their value for aid and good proxies for these values are available to those in charge of distribution. We find that targeting based on housing damages modestly increased the value of aid in this setting according to our measure of demand for liquidity. Targeting based on housing damages also promotes reconstruction, but it does not perform well by a utilitarian welfare criterion, which prioritizes low-wealth, low-consumption households.

Furthermore, we find that other targeting strategies would have generated more welfare by our measure of demand for liquidity. In particular, targeting entire villages that experienced the most intense shaking, or even a universal allocation, that gives a smaller amount of aid to all households in affected districts would have generated more welfare, although the latter result is sensitive to the estimated curvature of household value functions. These strategies have the additional benefit of not requiring any budget or time for conducting a needs assessment.

Several factors might affect the external validity of these conclusions. An earthquake is a particular type of disaster, in that the main effect is usually to destroy structures, which may be a less important input into production in a rural agricultural setting. Endogenizing the income process could be more important to analyze flooding or droughts, or targeting aid in an urban setting, and should be considered in future research. Second, the aid program analyzed here was delivered more than a year after the disaster. Although this time frame may be typical for reconstruction aid, the implications for targeting emergency relief in the immediate aftermath of a disaster could differ, especially when aid is in the form of in-kind goods rather than cash. Finally, analyzing the optimal amount of targeting likely depends upon the existence of other social insurance programs, which would be more important in wealthy and middle-income countries.

There are other potential downsides to allocating aid based on damages that are beyond the scope of this paper, but could be examined in future work. Aid that is conditional on property damage might create perverse incentives if property owners fail to internalize the full risks of building in disaster-prone areas, or under-invest in hazard mitigation more generally. This scenario is unlikely for a once-a-century earthquake in a country without major social insurance programs, but more plausible for recurrent disasters and more experience with post-disaster aid programs.

Furthermore, as mentioned in Section 2, disputes over beneficiary lists led to protests and delays. Given the importance of speed in aid delivery, more research should address the political economy concerns of targeting, and what types of allocations are easiest to administer. One possible explanation for the popularity of damage-based aid is that the eligibility criteria should be relatively transparent and easy to administer. This did not seem to be the case in Nepal, however ([The Asia Foundation, 2016b](#)).

Lastly, if aid has positive spillovers, or general equilibrium effects, as found by [Egger et al. \(2022\)](#), the value of precise targeting within villages is further diminished. Given these results, it is important to study whether alternative allocation mechanisms, including community-based or universal approaches, would have been perceived as more fair by those who lived through the disaster.

REFERENCES

- Aiken, Emily, Suzanne Bellue, Dean Karlan, Chris Udry, and Joshua E. Blumenstock.** 2022. “Machine learning and phone data can improve targeting of humanitarian aid.” *Nature*, 603(7903): 864–870. [3]
- Banerjee, Abhijit, Rema Hanna, Benjamin A Olken, and Diana Sverdlin Lisker.** 2023. “Social Protection in the Developing World.” *Working Paper*. [3]
- Basurto, Maria Pia, Pascaline Dupas, and Jonathan Robinson.** 2020. “Decentralization and efficiency of subsidy targeting: Evidence from chiefs in rural Malawi.” *Journal of Public Economics*, 185: 104047. [4]
- Bhusal, Bhishma, Michael Callen, Saad Gulzar, Rohini Pande, Soledad Artiz Prillaman, and Deepak Singhanian.** 2022. “Does Revolution Work? Evidence from Nepal’s People’s War.” *Working Paper*. [4, 7, 9]
- Billings, Stephen B., Emily A. Gallagher, and Lowell Ricketts.** 2022. “Let the rich be flooded: The distribution of financial aid and distress after hurricane harvey.” *Journal of Financial Economics*, 146(2): 797–819. [5]
- Botzen, W. J. Wouter, Olivier Deschenes, and Mark Sanders.** 2019. “The Economic Impacts of Natural Disasters: A Review of Models and Empirical Studies.” *Review of Environmental Economics and Policy*, 13(2): 167–188. [5]
- Calonico, Sebastian, Matias D Cattaneo, and Max H Farrell.** 2020. “Optimal bandwidth choice for robust bias-corrected inference in regression discontinuity designs.” *The Econometrics Journal*, 23(2): 192–210. [24, 27]
- Calonico, Sebastian, Matias D. Cattaneo, Max H. Farrell, and Rocío Titiunik.** 2017. “Rdrobust: Software for Regression-discontinuity Designs.” *The Stata Journal: Promoting communications on statistics and Stata*, 17(2): 372–404. [24]
- Cameron, Lisa, and Manisha Shah.** 2015. “Risk-Taking Behavior in the Wake of Natural Disasters.” *The Journal of Human Resources*, 50(2): 484–515. Publisher: [University of Wisconsin Press, Board of Regents of the University of Wisconsin System]. [30]

- Cattaneo, Matias D., Nicolás Idrobo, and Rocío Titiunik.** 2020. *A Practical Introduction to Regression Discontinuity Designs: Foundations*. Cambridge University Press. [24]
- Chetty, Raj.** 2006. “A general formula for the optimal level of social insurance.” *Journal of Public Economics*, 90(10-11): 1879–1901. [9]
- Deaton, Angus.** 1991. “Saving and Liquidity Constraints.” *Econometrica*, 59(5): 1221–1248. [16]
- Dell, Melissa, Benjamin F. Jones, and Benjamin A. Olken.** 2014. “What Do We Learn from the Weather? The New Climate-Economy Literature.” *Journal of Economic Literature*, 52(3): 740–798. [5]
- Deryugina, Tatyana.** 2017. “The Fiscal Cost of Hurricanes: Disaster Aid versus Social Insurance.” *American Economic Journal: Economic Policy*, 9(3): 168–198. Publisher: American Economic Association. [5]
- Deryugina, Tatyana, Laura Kawano, and Steven Levitt.** 2018. “The Economic Impact of Hurricane Katrina on Its Victims: Evidence from Individual Tax Returns.” *American Economic Journal: Applied Economics*, 10(2): 202–233. [5]
- Egger, Dennis, Johannes Haushofer, Edward Miguel, Paul Niehaus, and Michael Walker.** 2022. “General Equilibrium Effects of Cash Transfers: Experimental Evidence From Kenya.” *Econometrica*, 90(6): 2603–2643.
_eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA17945>. [42]
- FEMA.** 2022. “OpenFEMA Dataset: Individuals and Households Program - Valid Registrations - v1.” [2]
- Fenichel, Eli P., and Joshua K. Abbott.** 2014. “Heterogeneity and the fragility of the first best: Putting the “micro” in bioeconomic models of recreational resources.” *Resource and Energy Economics*, 36(2): 351–369. [14]
- Fu, Chao, and Jesse Gregory.** 2019. “Estimation of an Equilibrium Model With Externalities: Post-Disaster Neighborhood Rebuilding.” *Econometrica*, 87(2): 387–421. [4, 19]
- Gallagher, Justin, and Daniel Hartley.** 2017. “Household Finance after a Natural Disaster: The Case of Hurricane Katrina.” *American Economic Journal: Economic Policy*, 9(3): 199–228. [5]
- Gallagher, Justin, Daniel Hartley, and Shawn Rohlin.** 2023. “Weathering an Unexpected Financial Shock: The Role of Federal Disaster Assistance on Household Finance and Business Survival.” *Journal of the Association of Environmental and Resource Economists*, 10(2): 525–567. Publisher: The University of Chicago Press. [5]
- Government of Nepal National Planning Commission.** 2015. “Nepal Earthquake 2015 Post Disaster Needs Assessment.” Kathmandu. [7]
- Gregory, Jesse.** 2017. “The Impact of Post-Katrina Rebuilding Grants on the Resettlement Choices of New Orleans Homeowners.” [4]
- Gurung, Harka B., Yogendra Gurung, and Chhabi Lal Chidi.** 2006. *Nepal atlas of ethnic & caste groups*. Lalitpur: National Foundation for Development of Indigenous Nationalities. [26]
- Hallegatte, Stephane, Mook Bangalore, Laura Bonzanigo, Marianne Fay, Tamaro Kane, Ulf Narloch, Julie Rozenberg, David Treguer, and Adrien Vogt-Schilb.** 2016. *Shock Waves*. Washington, DC: World Bank. [3]
- Hanna, R., and D. Karlan.** 2017. “Chapter 7 - Designing Social Protection Programs: Using Theory and Experimentation to Understand How to Help Combat Poverty.” In *Handbook of Economic Field Experiments*. Vol. 2 of *Handbook of Economic Field Experiments*, , ed. Abhijit Vinayak Banerjee and Esther Duflo, 515–553. North-Holland. [3]

- Hanna, Rema, and Benjamin A. Olken.** 2018. “Universal Basic Incomes versus Targeted Transfers: Anti-Poverty Programs in Developing Countries.” *Journal of Economic Perspectives*, 32(4): 201–226. [3, 12, 36]
- Hansen, Lars Peter.** 1982. “Large Sample Properties of Generalized Method of Moments Estimators.” *Econometrica*, 50(4): 1029–1054. [22]
- Housing Recovery and Reconstruction Platform.** 2018. “Nepal, Gorkha earthquake: 18 moderately affected districts.” Housing Recovery and Reconstruction Platform. [7, 8, 23]
- ICRC.** 2025. “Family tent, ridge - Standard products catalogue IFRC ICRC.” [30]
- Imbens, Guido W., and Thomas Lemieux.** 2008. “Regression discontinuity designs: A guide to practice.” *Journal of Econometrics*, 142(2): 615–635. [26]
- Kaboski, Joseph P., and Robert M. Townsend.** 2011. “A Structural Evaluation of a Large-Scale Quasi-Experimental Microfinance Initiative.” *Econometrica*, 79(5): 1357–1406. [4, 16, 17, 20, 21, 29, 30, 39]
- Kansikas, Carolina, Anandi Mani, and Paul Niehaus.** 2025. “Structuring cash transfers: cash flow preferences, seasonality, and financial decisions in rural Kenya.” *NBER Working Paper*. [19]
- Keele, Luke J., and Rocío Titiunik.** 2015. “Geographic Boundaries as Regression Discontinuities.” *Political Analysis*, 23(1): 127–155. [26]
- Kellenberg, Derek, and A. Mushfiq Mobarak.** 2011. “The Economics of Natural Disasters.” *Annual Review of Resource Economics*, 3(1): 297–312. [5]
- Kolesár, Michal, and Christoph Rothe.** 2018. “Inference in Regression Discontinuity Designs with a Discrete Running Variable.” *American Economic Review*, 108(8): 2277–2304. [25]
- Mahadevan, Meera, and Ajay Shenoy.** 2023. “The political consequences of resource scarcity: Targeted spending in a water-stressed democracy.” *Journal of Public Economics*, 220: 104842. [4]
- McCrary, Justin.** 2008. “Manipulation of the running variable in the regression discontinuity design: A density test.” *Journal of Econometrics*, 142(2): 698–714. [26]
- Morten, Melanie.** 2019. “Temporary Migration and Endogenous Risk Sharing in Village India.” *Journal of Political Economy*, 127(1): 1–46. Publisher: The University of Chicago Press. [12]
- Munshi, Kaivan, and Mark Rosenzweig.** 2016. “Networks and Misallocation: Insurance, Migration, and the Rural-Urban Wage Gap.” *American Economic Review*, 106(1): 46–98. [12]
- Negishi, Takashi.** 1960. “Welfare Economics and Existence of an Equilibrium for a Competitive Economy.” *Metroeconomica*, 12(2-3): 92–97. [14]
- Nepal Earthquake Housing Reconstruction Program Multi Donor Trust Fund.** 2016. “Program Overview and Operations Manual.” Nepal Earthquake Housing Reconstruction Program Multi Donor Trust Fund. [7]
- Niehaus, Paul, Antonia Atanassova, Marianne Bertrand, and Sendhil Mullainathan.** 2013. “Targeting with Agents.” *American Economic Journal: Economic Policy*, 5(1): 206–238. [3]
- Nordhaus, William D., and Zili Yang.** 1996. “A Regional Dynamic General-Equilibrium Model of Alternative Climate-Change Strategies.” *The American Economic Review*, 86(4): 741–765. Publisher: American Economic Association. [14]
- Pathak, Prakash, and Matthias Schündeln.** 2022. “Social hierarchies and the allocation of development aid: Evidence from the 2015 earthquake in Nepal.” *Journal of Public Economics*, 209: 104607. [4, 7]

- Randell, Heather, Chengsheng Jiang, Xin-Zhong Liang, Raghu Murtugudde, and Amir Sapkota.** 2021. "Food insecurity and compound environmental shocks in Nepal: Implications for a changing climate." *World Development*, 145: 105511. [6]
- Strömberg, David.** 2007. "Natural Disasters, Economic Development, and Humanitarian Aid." *Journal of Economic Perspectives*, 21(3): 199–222. [15]
- Tarquinio, Lisa.** 2022. "The Politics of Drought Relief: Evidence from Southern India." 73. [4, 5]
- The Asia Foundation.** 2016a. "Independent Impacts and Recovery Monitoring Phase 3." 106. [7, 20]
- The Asia Foundation.** 2016b. "Nepal Government Distribution of Earthquake Reconstruction Cash Grants for Private Houses." [2, 6, 7, 42]
- Townsend, Robert M.** 1994. "Risk and Insurance in Village India." *Econometrica*, 62(3): 539–591. [5, 9]
- Walker, Thomas, Yasuhiro Kawasoe, and Jui Shrestha.** 2019. *Risk and Vulnerability in Nepal*. World Bank, Washington, DC. [7]
- Yang, Fang.** 2009. "Consumption over the life cycle: How different is housing?" *Review of Economic Dynamics*, 12(3): 423–443. [16]

ONLINE APPENDIX: TARGETING DISASTER AID: A STRUCTURAL
EVALUATION OF A LARGE EARTHQUAKE RECONSTRUCTION
PROGRAM

MATTHEW D. GORDON

Department of Economics, Paris School of Economics

YUKIKO HASHIDA

Agricultural and Applied Economics, University of Georgia

ELI P. FENICHEL

School of the Environment, Yale University

APPENDIX A: APPENDIX A. STRUCTURAL MODEL AND ESTIMATION

A.1. Value Function Normalization

Define $X_{it}, H_{it}, C_{it}, I_{it}, Y_{it}$ as the true values of wealth, housing, consumption, housing investment, and income respectively, with $x_{it}, h_{it}, c_{it}, \iota_{it}, y_{it}$ as the same variables normalized by $E(Y_{it}) = P_{it} = \exp(\mu_{it} + \sigma^2/2)$. From equation (5) we have:

$$\begin{aligned}
 U_i &= E \left[\sum_{t=0}^{\infty} \beta^t \frac{C_{it}^\alpha H_{it}^{1-\alpha} (1-\gamma)}{1-\gamma} \right] \\
 &= E \left[\sum_{t=0}^{\infty} \beta^t \frac{((P_{it} c_{it})^\alpha (P_{it} h_{it}))^{(1-\alpha)}}{1-\gamma} (1-\gamma) \right] \\
 &= P_{it}^{1-\gamma} E \left[\sum_{t=0}^{\infty} \beta^t \frac{c_{it}^\alpha h_{it}^{1-\alpha} (1-\gamma)}{1-\gamma} \right].
 \end{aligned} \tag{15}$$

Thus if ν is the value function associated with maximizing $E[\sum_{t=0}^{\infty} \beta^t u(c_{it}, h_{it})]$, $V_i = P_{it}^{1-\gamma} \nu$ is the value function associated with maximizing equation (5). The normalization carries through to all the state variables without issue, observing that $\log(y_{it}) = \log(Y_{it}/P_{it}) = \log(Y_{it}) - \log(P_{it}) = \log(Y_{it}) - \mu_{it} - \frac{\sigma^2}{2} \sim N(-\frac{\sigma^2}{2}, \sigma^2)$.

Therefore, we solve for:

$$\begin{aligned}
 P_{it}^{\gamma-1} V(x_{it}, h_{it} | P_{it}) &= \nu(x_{it}, h_{it}) = \max_{D_{it}} \left\{ \right. \\
 &D_{it} \left[u(\bar{c}, \max[h_{it}, \bar{h}], 0) + \beta E[\nu(R\lambda + y_{it+1}, \delta \max[h_{it}, \bar{h}])] \right], \\
 &\left. (1 - D_{it}) \left[\max_{c_{it}, \iota_{it}} u(c_{it}, h_{it}, \iota_{it}) + \beta E[\nu(R(x_{it} - c_{it} - \iota_{it}) + y_{it+1}, \delta h_{it} + \iota_{it+1})] \right] \right\}
 \end{aligned} \tag{16}$$

subject to the constraints:

$$x_{it} - c_{it} - \iota_{it} \geq \lambda \tag{17}$$

$$\iota_{it} \geq 0. \tag{18}$$

and the corresponding optimal consumption and investment functions.

The closed form expression for τ comes the definition of τ as:

$$\begin{aligned}
 V_i(x_{it}, h_{it}, P_{it}) &= V_i(x_{it} + a, h_{it}, (1 - \tau_{it})P_{it}) \\
 P_{it}^{1-\gamma} \nu_i(x_{it}, h_{it}) &= P_{it}^{1-\gamma} (1 - \tau)^{1-\gamma} \nu_i(x_{it} + a, h_{it}) \\
 \frac{\nu_i(x_{it}, h_{it})}{\nu_i(x_{it} + a, h_{it})} &= (1 - \tau)^{1-\gamma} \\
 \tau &= 1 - \left[\frac{\nu_i(x_{it} + a, h_{it})}{\nu_i(x_{it}, h_{it})} \right]^{1/(\gamma-1)}
 \end{aligned} \tag{19}$$

Since the value function is negative everywhere and bounded by 0, the ratio is < 1 , and the ratio is decreasing in the increase in utility from aid, so τ is increasing.

A.2. Finding the Policy Functions

It is possible to solve for the value function and optimal policy functions, $c^*(x, h, P)$, $\iota^*(x, h, P)$, and $D^*(x, h, P)$, given a set of parameters using Value Function Iteration (VFI). We start with the guess that the value function is equal to zero everywhere. The value function is stored in a grid of 1600 points, with 40 equally-spaced points in both the x and h dimensions. We interpolate the value function between grid points using the multidimensional simplicial scheme described in [Judd \(1998\)](#). The income process is discretized by exponentiating 30 Gaussian quadrature points. At each grid point we find the values of c and ι that maximize A1. Since the default decision introduces a kink in the value function, which can lead to local minima in the objective function resulting from the approximation and interpolation scheme, we use a semi-global optimization algorithm at each grid point ([Gablonsky and Kelley, 2001](#), [Johnson, 2022](#)).

The value of the objective function at the maximum is then stored as ν_2 . Thus, for each subsequent iteration $n+1$, we solve for:

$$\nu_{n+1} = \max u(c_t, h_t) + \beta E[\nu_n(x_{t+1}, h_{t+1})]. \tag{20}$$

This is repeated until the relative mean squared difference in $c^*(x, h)$ between iterations is less than .0025.

A.3. Parameter Estimation

The parameter vector is estimated by minimizing the MSM objective function using a set of 17 moment conditions derived from the model described below. To do so we use a nested fixed-point algorithm: the outer loop iterates over parameters, while the inner loop estimates the optimal policy functions at each set of parameters.

Since approximation errors in the inner loop can generate local minima in the objective function, we first search for an approximate global minima of equation (21) using the DIRECT global optimization algorithm (Powell, 1998). We use that minima to estimate the optimal weighting matrix, W (Hansen, 1982), and also use it as the starting point of a local optimization algorithm in a final iteration to estimate the parameter vector ($\hat{\theta}$) to satisfy:

$$\hat{\theta} = \arg \min_{\theta} \left(\sum_{i=1}^N \sum_{t=2}^3 \sqrt{w_i} g_{it}(\theta, x_{it}, h_{it}, P_{it}, \iota_{it}, c_{it}, D_{it}, y_{it}) \right)^T \widehat{W} \left(\sum_{i=1}^N \sum_{t=2}^3 \sqrt{w_i} g_{it}(\theta, x_{it}, h_{it}, P_{it}, \iota_{it}, c_{it}, D_{it}, y_{it}) \right). \quad (21)$$

Where $g_{it} = \{e_{it1}, \dots, e_{it17}\}^T$, w_i are the survey weights, and $\widehat{W}$ is the inverse of the estimated covariance matrix of moments from the first stage optimization. The first survey wave is dropped because lagged cash savings and lagged housing value are needed to construct the state variables.

A.3.1. Moment Conditions

The optimal policy functions for consumption, investment, and default define the first three moment conditions:

$$\begin{aligned} e_{it1} &= c^*(x_{it}, h_{it}, P_{it}) - c_{it} \\ e_{it2} &= \iota^*(x_{it}, h_{it}, P_{it}) - \iota_{it} \\ e_{it3} &= D^*(x_{it}, h_{it}, P_{it}) - D_{it}. \end{aligned} \quad (22)$$

Following Kaboski and Townsend (2011), we gain six additional moment conditions by interacting e_1 and e_2 with transformations of each of the state variables. Intuitively, this

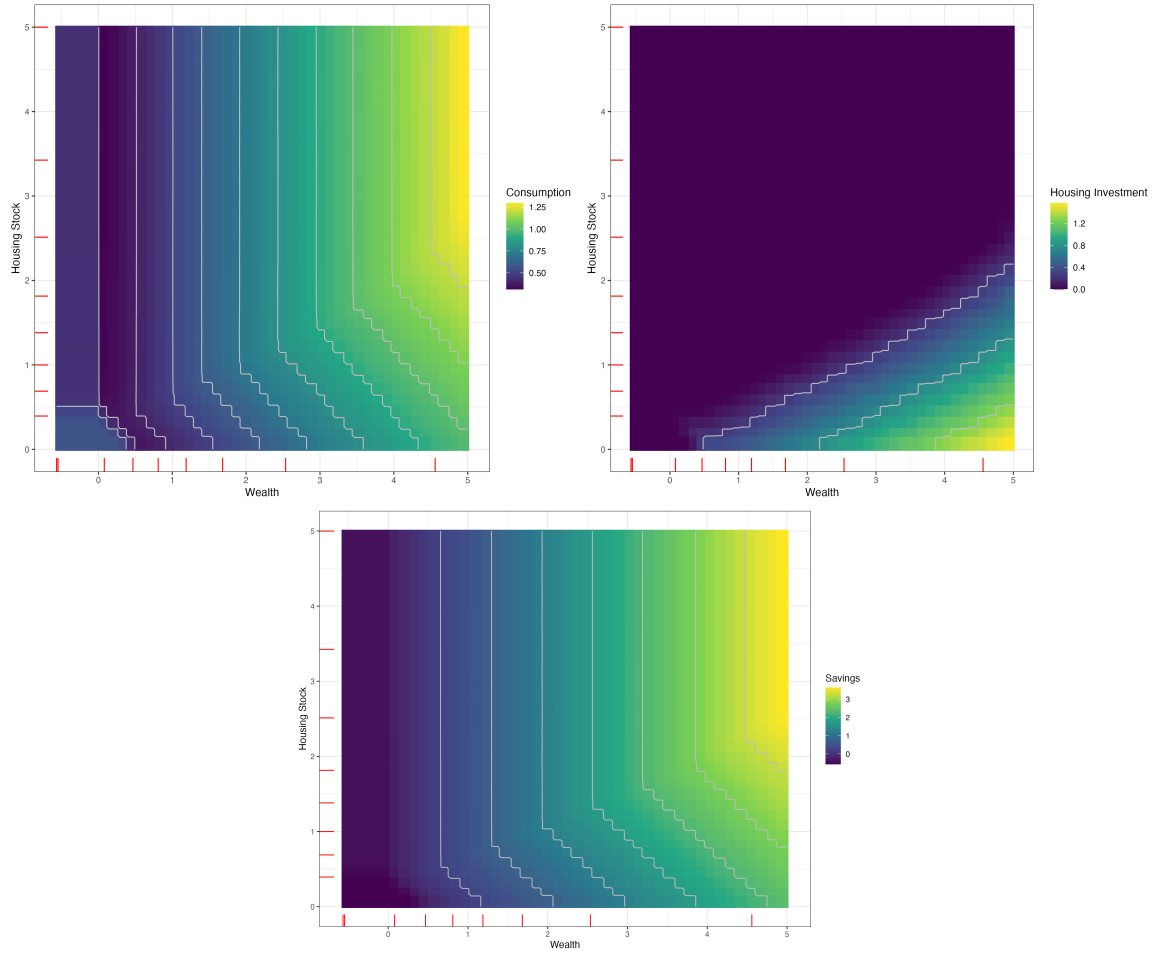


FIGURE A.1.—Estimated Policy Functions for consumption (top left), housing investment (top right), and savings (bottom). Scale is fraction of expected income. Red lines in margins show 10-90th deciles of households in the data.

helps ensure that the model's predictions are not biased in expectation for any values of the state variables:

$$e_{it4} = e_{it1} \log(P_{it})$$

$$e_{it5} = e_{it1} \text{ ihs}(x_{it})$$

$$e_{it6} = e_{it1} \log(h_{it} + 1)$$

$$e_{it7} = e_{it2} \log(P_{it})$$

$$e_{it8} = e_{it2} \text{ ihs}(x_{it})$$

$$e_{it9} = e_{it2} \log(h_{it} + 1).$$

The inverse hyperbolic sine function is used for x_{it} to handle negative values.

Moment conditions 10-12 help to pin down interest rates by using data on household capital income and loan payments made and received.

$$\begin{aligned} e_{it10} &= (R - 1) \text{Cash Savings}_{it} - \text{Capital Gains}_{it} \\ e_{it11} &= \sum_{T=1}^3 \frac{R^T(R - 1)}{R^T - 1} \text{Loans Taken}_{it} - \text{Loan Payments Made}_{it} \\ e_{it12} &= \sum_{T=1}^3 \frac{R^T(R - 1)}{R^T - 1} \text{Loans Made}_{it} - \text{Loan Payments Received}_{it} \end{aligned}$$

T is the term of the loan. We use any outstanding consumption (non-investment) loans in period t with a 3 year term or less.

The next moment condition helps to pin down depreciation rates based on a hedonic approach using the age of the home. Consider the following regression of home values (unnormalized) on home investment and the age of the home with household and year fixed effects:

$$\log H_{it} = a_1 \text{Age of Home}_{it} + a_2 \text{Housing Investment}_{it} + \eta_i + \phi_t + e_{it}. \quad (23)$$

The coefficient $a_1 = \delta - 1$ in the model. We can use the Frisch-Waugh-Lovell theorem to derive an appropriate moment condition based on this regression:

$$e_{it13} = \widetilde{\text{Age of Home}}_{it} \left(\widetilde{H}_{it} - (\delta - 1) \widetilde{\text{Age of Home}}_{it} \right).$$

where $\widetilde{H}_{it}$ and $\widetilde{\text{Age of Home}}_{it}$ are the residuals of a regression of those variables on housing investment and the household and survey year fixed effects. Table B.V shows estimates of equation 23, as well as robustness of the estimates to other specifications.

The final two moment conditions helps identify the variance of the income process and measurement error. In the simulated data, we draw 100 measurement error shocks for each observation from $\log(s_{it}) \sim N(\frac{\sigma_m^2}{2}, \sigma_m^2)$. We then multiply P_{it} by these shocks, which cre-

ates measurement error in liquid wealth and housing wealth when normalizing those variables by permanent income.

This distribution ensures that for any state variable q_{it} , $E(\frac{q_{it}}{P_{it}s_{it}}) = \frac{q_{it}}{P_{it}}E(1/s_{it}) = \frac{q_{it}}{P_{it}}e^{-\frac{\sigma_m^2}{2} + \frac{\sigma_m^2}{2}} = \frac{q_{it}}{P_{it}}$, so all previous moment conditions remain valid in their normalized form.

Then $\log(\tilde{y}_{it}) := \log(Y_{it}/(P_{it}s_{it})) \sim N(-\frac{\sigma^2}{2} - \frac{\sigma_m^2}{2}, \sigma^2 + \sigma_m^2)$. Therefore we use the following moment conditions:

$$\begin{aligned} e_{it14} &= \log(\tilde{y}_{it}) + \frac{\sigma^2}{2} + \frac{\sigma_m^2}{2} \\ e_{it15} &= \log(\tilde{y}_{it})^2 - \left(\frac{\sigma^2}{2} + \frac{\sigma_m^2}{2}\right)^2 - \sigma^2 - \sigma_m^2. \end{aligned}$$

All moment conditions are normalized by dividing by the average value of the relevant variable in the data (e.g. average consumption, investment, default, etc...). We estimate these moments using 11,115 household-year observations from the second and third waves – we also drop approximately 5% of households that did not answer large portions of the survey, making the construction of key variables impossible. For households that answered ‘Don’t Know’ for certain rare types of income, especially capital gains, we impute a zero for that category of income. In estimating standard errors for the parameters, we cluster by household.

A.3.2. *Purging Lifecycle Variation and Household Heterogeneity*

Since the data contain variation not explicitly modelled, including life-cycle considerations and other unobserved determinants of household heterogeneity, we follow the buffer stock literature in purging these sources of variation from the estimation procedure (Kamboski and Townsend, 2011). This procedure is necessary to ensure that household values for aid are not biased by life cycle considerations, household size, or other systematic differences between households. Consumption and savings patterns may differ between older and younger families, for example, even if they had the same values for aid. Purging variation associated with these differences requires careful consideration, however, since our counterfactuals address targeting, the value of which depends on household heterogeneity.

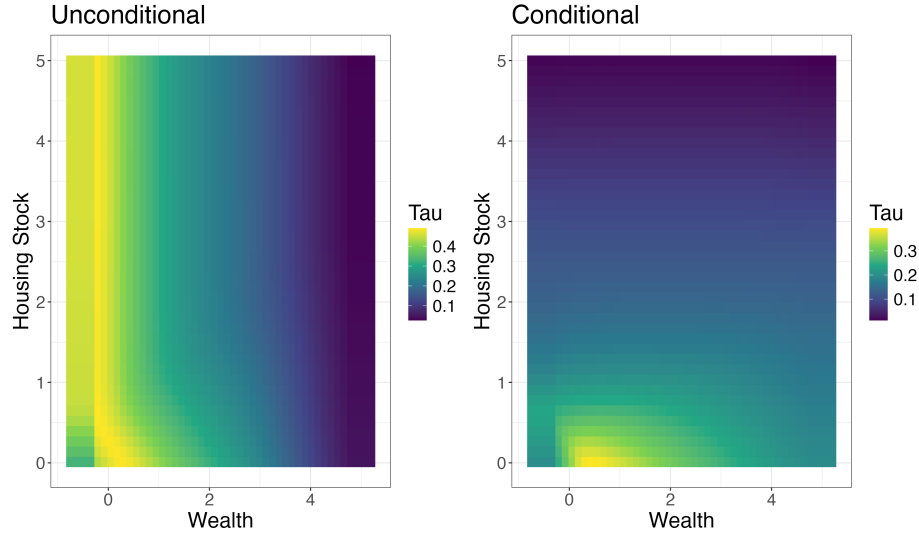


FIGURE A.2.—Willingness to Pay for Aid as a function of state variables. Tau represents WTP as a fraction of expected income. WTP is calculated using equation 3. Left shows WTP for unconditional aid, right show WTP for aid that must be invested in housing.

To be precise, our model says that households are only heterogeneous in their history of shocks to income and housing stock – including the earthquake – and their expected income, and we seek to understand how this heterogeneity correlates with different targeting programs.

If a targeting program correlates with any of the exogenous characteristics purged from the data, then removing that source of variation will remove any value created (or destroyed) by systematically targeting those households. Thus purging regional variation, for example, is undesirable, since earthquake damages vary across space, and a targeting program might want to take that into account.

On the other hand, the age structure of the household, household size (including migrant members), and education can be plausibly seen as exogenous to the earthquake damages, as well as the targeting strategies under consideration, yet may account for systematic differences in behavior. Survey year fixed effects are included to capture macroeconomic fluctuations such as price changes and exchange rates.

Purging these sources of variation results in a nuanced interpretation of household values for aid. The estimates reflect household value for aid conditional on the set of exogenous household characteristics. This is consistent with the idea that aid is to be used for

smoothing welfare through the disaster, and not redistributing between different types of households.

Thus we run the following regressions:

$$\begin{aligned}
 \log(C_{it} + 1) &= \Gamma_1 W_{it} + \epsilon_{it}^1 \\
 \log(H_{it} + 1) &= \Gamma_2 W_{it} + \epsilon_{it}^2 \\
 D_{it} &= \Gamma_3 W_{it} + \epsilon_{it}^3 \\
 X_{it} &= \Gamma_4 W_{it} + \epsilon_{it}^4
 \end{aligned} \tag{24}$$

Where W_{it} is a vector of household characteristics containing quadratic polynomials of age of the head of household, education, and the number of members (including migrants), as well as the number of children and elderly members, and survey wave fixed effects. We add one to consumption and housing to maintain households with a zero value for those variables, since these households could be particularly valuable from a targeting perspective. The R squared values of these regressions are .32, .07, .08, and .01 respectively.

We then construct an adjusted dataset, where the values of consumption, housing wealth, default, and liquidity are the fitted values of 24 for a household with mean values of the independent variables, plus the household specific residual. For example, adjusted household consumption is constructed as:

$$\tilde{C}_{it} = \exp(\hat{\Gamma}_1 \bar{W}_{it} + \hat{\epsilon}_{it}^1). \tag{25}$$

Income is treated similarly, but we also allow for additional household heterogeneity in order to more accurately estimate expected income. We include a vector of additional characteristics, U_{it} , which includes the value of household landholdings, the gender of the head of household, the number and destination of previous migrants, migrant earnings, ethnicity and village fixed effects. The R squared of the income regression with these additional characteristics is .24. We then construct expected income and income shocks as follows:

$$\log(Y_{it} + 1) = \Gamma_5 W_{it} + \omega U_{it} + \epsilon_{it}^5 \tag{26}$$

$$\tilde{Y}_{it} = \exp(\hat{\Gamma}_5 \bar{W}_{it} + \hat{\omega} U_{it} + \hat{\epsilon}_{it}^5) \quad (27)$$

$$P_{it} = \exp(\hat{\Gamma}_5 \bar{W}_{it} + \hat{\omega} U_{it}) \hat{\xi}_t \quad (28)$$

Where $\hat{\xi}_t = \sum_i w_i \exp(\hat{\epsilon}_{it}^5)$ is the weighted smear factor used to retransform the expected income values back to the scale of the original variable (Duan, 1983). The adjusted values of consumption, housing, default, liquidity, and income are all also multiplied by a common factor to ensure that their means in the adjusted data are the same as in the original data. Lastly, housing investment is adjusted by multiplying housing investment in the data by the ratio of adjusted income to unadjusted income. The adjusted data is then used to both estimate the parameters and conduct counterfactuals.

All code is written in R and is available from:

https://github.com/mdgordo/nepal_earthquake/.

APPENDIX REFERENCES:

- Duan, Naihua.** 1983. “Smearing Estimate: A Nonparametric Retransformation Method.” *Journal of the American Statistical Association*, 78(383): 605–610. [55]
- Funk, Chris, Pete Peterson, Martin Landsfeld, Diego Pedreros, James Verdin, Shraddhanand Shukla, Gregory Husak, James Rowland, Laura Harrison, Andrew Hoell, and Joel Michaelsen.** 2015. “The climate hazards infrared precipitation with stations—a new environmental record for monitoring extremes.” *Scientific Data*, 2(1): 150066. Publisher: Nature Publishing Group. [60]
- Gablonsky, J.M., and C.T. Kelley.** 2001. “A Locally-Biased form of the DIRECT Algorithm.” *Journal of Global Optimization*, 21(1): 27–37. [48]
- Hansen, Lars Peter.** 1982. “Large Sample Properties of Generalized Method of Moments Estimators.” *Econometrica*, 50(4): 1029–1054. [49]
- Johnson, Steven G.** 2022. “stevengj/nlopt.” original-date: 2013-08-27T16:59:11Z. [48]
- Judd, Kenneth L.** 1998. *Numerical Methods in Economics*. Cambridge, MA, USA:MIT Press. [48]
- Kaboski, Joseph P., and Robert M. Townsend.** 2011. “A Structural Evaluation of a Large-Scale Quasi-Experimental Microfinance Initiative.” *Econometrica*, 79(5): 1357–1406. [49, 52]
- Powell, M. J. D.** 1998. “Direct search algorithms for optimization calculations.” *Acta Numerica*, 7: 287–336. Publisher: Cambridge University Press. [49]

APPENDIX B: ADDITIONAL FIGURES AND TABLES

B.1. *Earthquake Aid, Nepal Timeline, and Summary Statistics*

Earthquake	Overview of Aid Response and Cash Transfers
Myanmar/Thailand (2025)	\$275 million flash appeal. Small cash grants (\$86-172) targeted broadly in earthquake affected areas given numerous ongoing humanitarian crises (UN Myanmar 2025: <i>Myanmar Earthquake: HNRP Flash Addendum</i>)
Afghanistan (2025, 2023, 2022)	Afghanistan suffered earthquakes in 2022, 2023, and 2025, while contending with droughts, flooding, and conflict. The \$173 million aid request after the Herat earthquakes called for \$73.3 million in funding for emergency shelter, mostly in-kind. Cash transfers were not a major component. (OCHA 2023: <i>Afghanistan Revised Herat Earthquake Response Plan</i>).
Morocco (2023)	\$11.7 billion reconstruction fund provided up to \$15,000 per household in cash grants conditional on earthquake damages, among other expenditures (Reuters 2023: <i>Morocco to spend \$11.7 billion on five-year post-quake reconstruction plan</i>).
Türkiye/Syria (2023)	\$1 billion fund provided \$530 payments to households conditional on earthquake damages. Additional rental assistance and relocation support grants as well. (UNHCR 2023: <i>Cash Assistance Needs and Efforts in the Context of the Earthquake Response in Türkiye</i>).
Haiti (2021)	\$77 million in funding received, cash grants not-prioritized, mostly in-kind food and shelter assistance as well as funding for the education sector (OCHA 2021: <i>Haiti: Earthquake Situation Report No. 3</i>).
Indonesia (2018)	Government grants of \$700-3,500 conditional on degree of earthquake damages (Hidayah, Nur et al 2020: <i>Housing rehabilitation and reconstruction in Central Sulawesi post-2018 earthquake</i> . E3S Web of Conferences).
Nepal (2015)	\$4.1 billion multilateral reconstruction fund, largest component was cash grants to households up to \$3,000 conditional on earthquake damages (see main text).
Haiti (2010)	\$6 billion in aid disbursed, channeled through NGOs and private entities. Lack of transparency made it difficult to ascertain how funds were spent—few NGOs published any cost breakdowns of spending (Ramachandran Vijaya and Julie Walz (2012) <i>Haiti: Where Has All the Money Gone?</i> CGD Policy Paper 004.)

TABLE B.I

RECENT EARTHQUAKES IN LOW AND MIDDLE-INCOME COUNTRIES

Description of aid and cash transfer programs after earthquakes 2010-2025 with greater than 1,000 deaths.

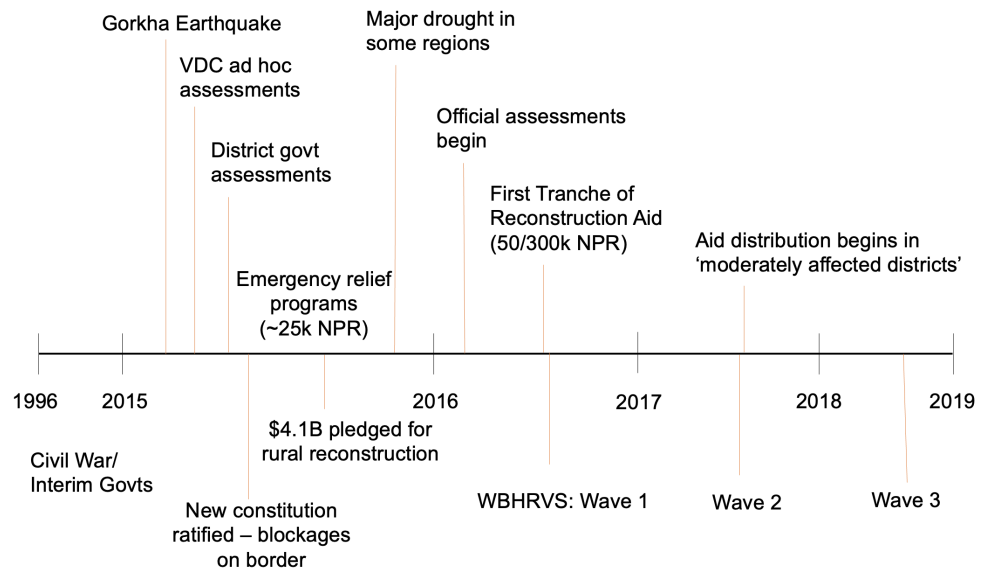


FIGURE B.1.—Timeline of Events

Statistic	N	Mean	Std.Dev.	Min	Max
Consumption	17,782	153,227	89,471	10,838	3,908,772
Food Consumption	17,782	114,077	60,864	1,560	938,704
Income	17,782	275,265	1,831,704	196	220,557,220
Productive Assets	17,779	1,946,683	10,007,589	0	809,395,500
Investments	17,780	39,110	284,253	0	15,000,000
Jewelery	17,742	73,040	174,598	0	12,500,000
School Costs	17,773	14,941	43,174	0	4,477,000
Home Value	17,781	1,228,095	2,232,228	0	70,000,000
Home Investment	17,782	23,083	154,747	0	10,500,000
Age of House	17,782	15.3	13.6	0	122
Household Members	17,478	4.66	1.97	1	21
Connected Migrants	17,782	0.77	1.19	0	18
Migrants Past Year	17,782	0.23	0.63	0	11
Overseas Migrants	17,782	0.40	0.71	0	8
Female Headed(%)	17,478	0.21	0.41	0	1
Cash Savings	17,708	50,444	160,758	0	5,015,000
Remittance Income	17,782	63,000	164,267	0	5,000,000
Loans Taken Past Year	17,768	28,726	123,351	0	7,000,000
Loans Made Past Year	17,782	2,198	26,278	0	1,500,000
Default(%)	17,757	0.19	0.40	0	1
Skipped Meal(%)	17,782	0.10	0.31	0	1
Earthquake Aid	17,782	18,440	74,826	0	2,182,000
Earthquake Aid(%)	17,782	0.13	0.37	0	1
NGO Aid	17,782	830	15,405	0	700,000
Public Transfers	17,736	4,482	10,875	0	320,500
Informal Transfers	17,782	2,571	22,804	0	2,807,540
Earthquake Losses (%)	17,782	0.21	0.43	0	1
Earthquake Losses	17,782	47,436	200,454	0	4,000,000

TABLE B.II

SUMMARY STATISTICS

Consumption includes value of all food consumption, energy, utilities, transportation and miscellaneous purchases, but not durables. Income is the sum of wages, rental income, agriculture and livestock sales, home food production, business revenues, pension and other public welfare. It does not include remittances or transfers. Productive Assets are the sum of the value of land, agricultural equipment, and livestock. Investments include land purchased, business investments, livestock purchases or farm equipment purchases. Cash savings includes cash on hand and bank savings plus insurance and savings group assets. Loans reflect short term credit (three years or less). Survey weights are used to calculate means and standard deviations. N is non-missing household-year observations in the three year panel.

	log(consumption)				
	(1)	(2)	(3)	(4)	(5)
log(income)	0.107*** (0.012)	0.113*** (0.017)	0.106*** (0.012)	0.116*** (0.010)	0.104*** (0.011)
log(income):Dalit		-0.017 (0.024)			
log(income):Newar		-0.053* (0.028)			
log(income):Other		-0.005 (0.017)			
log(income):Female Head			-0.010*** (0.002)		
log(income):Land > Median				-0.018*** (0.005)	
log(income):Quake Affected					0.015 (0.019)
Observations	16 742	16 741	16 742	16 742	16 742
R2	0.727	0.727	0.728	0.727	0.727

TABLE B.III

CONSUMPTION SMOOTHING: CASTE YEAR FIXED EFFECTS

All regressions include household and caste-year fixed effects. Standard errors clustered at the survey strata. Households with zero or missing income, or less than 3 observations dropped. Land percentile defined based on household land value during first survey wave. Omitted category for caste is "Brahmin/Chhetri". * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	log(consumption)				
	(1)	(2)	(3)	(4)	(5)
fit log(income)	0.274*** (0.018)	0.217*** (0.023)	0.271*** (0.017)	0.272*** (0.024)	0.271*** (0.020)
fit log(income):Dalit		0.082*** (0.024)			
fit log(income):Newar		−0.041 (0.065)			
fit log(income):Other		0.076*** (0.020)			
fit log(income):Female Head			0.000 (0.002)		
fit log(income):Land > Median				−0.012 (0.035)	
fit log(income):Quake Affected					−0.194*** (0.037)
Observations	16 742	16 741	16 742	16 742	16 742
R2	0.653	0.656	0.655	0.658	0.664

TABLE B.IV

CONSUMPTION SMOOTHING: RAINFALL INSTRUMENTS

All regressions include household fixed effects, and instrument for income using total rainfall in each of the preceding 12 months. Rainfall data comes from Funk et al. (2015). Standard errors clustered at the household. Households with zero or missing income, or less than 3 observations dropped. Land percentile defined based on household land value during first survey wave. Omitted category for caste is "Brahmin/Chhetri". * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	log(house value) (1)	log (house value + 1) (2)	house value (3)	log(house value) (4)	log (house value + 1) (5)	house value (6)	log(house value) (7)	log (house value + 1) (8)	house value (9)
Age of House	−0.020*** (0.005)	−0.022*** (0.006)	−0.018*** (0.005)	−0.017*** (0.005)	−0.017** (0.005)	−0.018*** (0.005)	−0.015** (0.005)	−0.016** (0.005)	−0.015*** (0.005)
Home Investment				0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Observations	16 591	16 742	16 742	16 591	16 742	16 742	16 591	16 742	16 742
R2	0.700	0.531		0.701	0.532		0.716	0.544	
Estimator	OLS	OLS	Poisson	OLS	OLS	Poisson	OLS	OLS	Poisson
Year FEs	X	X	X	X	X	X			
Linear Time Trend							X	X	X
District*Year FEs							X	X	X

TABLE B.V

HOUSING DEPRECIATION: HEDONIC REGRESSIONS

Coefficients of OLS and Poisson regression analogs of the housing depreciation moment condition described in Appendix A. Columns differ in the fixed effects included and whether the dependent variable is log house value (with zeros dropped) or log house value + 1 in the OLS specifications, or house value for the poisson specifications. All regressions have household fixed effects and standard errors clustered at the survey strata. Households with less than 3 observations dropped. Column 5 is the basis of the moment condition estimated using MSM. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

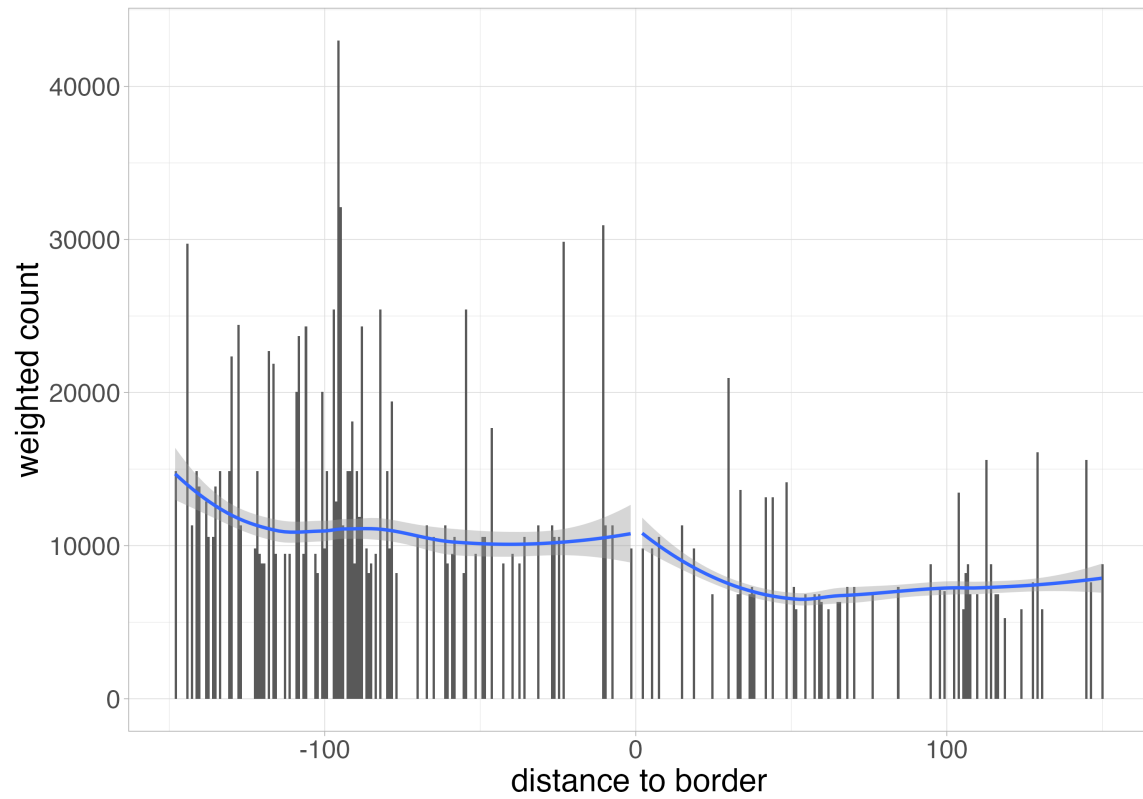
B.2. Regression Discontinuity Plots and Tables

FIGURE B.2.—McCrary Test: Weighted density of households on either side of the border with smoothed means.

	Received Aid (1)	Aid Amount (2)	Received Aid (3)	Aid Amount (4)	Received Aid (5)	Aid Amount (6)	Received Aid (7)	Aid Amount (8)	Received Aid (9)	Aid Amount (10)	Received Aid (11)	Aid Amount (12)
Distance > 0	0.23*** (0.07)	39,750*** (7,138)	0.18*** (0.06)	38,076*** (8,772)	0.17*** (0.06)	38,661*** (8,699)	0.27*** (0.07)	42,113*** (7,552)	0.53*** (0.10)	48,610*** (10,325)	0.26*** (0.08)	56,044*** (12,247)
N	1093	2449	1485	1006	1357	1051	784	1881	476	608	1053	1009
Bandwidth	35.43	62.13	43.01	33.56	40.2	33.77	28.53	50.97	20.39	23.57	36.5	35.7
Kernel	triangular	triangular	triangular	triangular	triangular	triangular	epanechnikov	epanechnikov	uniform	uniform	triangular	triangular
Controls			Damages	Damages	Damages/Geo/Demo	Damages/Geo/Demo						
Exclude unaffected Dists.												
5 km Donut												
Border	Western	Western	Western	Western	Western	Western	Western	Western	Western	Western	X Western	X Western
Wards	57	87	66	39	59	39	37	70	32	52	64	49

	Received Aid (1)	Aid Amount (2)	Received Aid (3)	Aid Amount (4)	Received Aid (5)	Aid Amount (6)	Received Aid (7)	Aid Amount (8)	Received Aid (9)	Aid Amount (10)	Received Aid (11)	Aid Amount (12)
Distance > 0	0.44*** (0.10)	42,112*** (11,293)	0.22*** (0.06)	39,354*** (8,292)	0.29*** (0.05)	40,542*** (7,224)	0.24*** (0.07)	39,688*** (7,146)	0.24*** (0.05)	16,635*** (3,885)	0.22*** (0.05)	12,550*** (3,729)
N	476	476	1357	1357	2360	2360	1093	2404	1580	3026	1450	3290
Bandwidth	20	20	40	40	60	60	34.66	62.07	10.55	23.65	10.41	24.64
Kernel	triangular	triangular	triangular	triangular	triangular	triangular	triangular	triangular	triangular	triangular	triangular	triangular
Controls							X	X			Damages/Geo/Demo	Damages/Geo/Demo
Exclude unaffected Dists.												
5 km Donut												
Border	Western	Western	Western	Western	Western	Western	Western	Western	Whole	Whole	Whole	Whole
Wards	11	11	32	32	56	56	54	81	65	112	65	115

	Received Aid (1)	Aid Amount (2)	Received Aid (3)	Aid Amount (4)	Received Aid (5)	Aid Amount (6)	Received Aid (7)	Aid Amount (8)	Received Aid (9)	Aid Amount (10)
Distance > 0	0.41*** (0.09)	62,630*** (18,672)	0.33*** (0.08)	89,547*** (28,797)	0.36*** (0.07)	88,240*** (28,087)	0.78*** (0.15)	45,729*** (13,650)	0.92 (656,518)	100,959*** (28,846)
N	1047	566	1355	215	1660	215	171	1181	126	215
Bandwidth	40.86	34.7	47.72	22.29	52	22.59	19.84	42.57	15.33	23.18
Kernel	triangular	triangular	triangular	triangular	triangular	triangular	epanechnikov	epanechnikov	uniform	uniform
Controls			Damages	Damages	Damages/Geo/Demo	Damages/Geo/Demo				
Exclude unaffected Dists.										
5 km Donut										
Border	Close Point	Close Point	Close Point	Close Point	Close Point	Close Point	Close Point	Close Point	Close Point	Close Point
Wards	57	61	64	32	69	34	32	70	21	57

	Received Aid (1)	Aid Amount (2)	Received Aid (3)	Aid Amount (4)	Received Aid (5)	Aid Amount (6)	Received Aid (7)	Aid Amount (8)	Received Aid (9)	Aid Amount (10)
Distance > 0	0.59*** (0.11)	71,027*** (19,821)	0.42*** (0.10)	52,012*** (15,376)	0.42*** (0.07)	44,312*** (11,215)	3.76*** (0.59)	66,627*** (19,863)	0.41*** (0.09)	57,931*** (17,357)
N	479	479	960	960	1880	1880	173	479	1047	829
Bandwidth	30	30	40	40	60	60	22.62	33.3	41.14	37.67
Kernel	triangular	triangular	triangular	triangular	triangular	triangular	triangular	triangular	triangular	triangular
Controls										
Exclude unaffected Dists.										
5 km Donut							X	X	X	X
Border	Close Point	Close Point	Close Point	Close Point	Close Point	Close Point	Close Point	Close Point	Close Point	Close Point
Wards	11	11	23	23	45	45	44	57	56	63

TABLE B.VI

FIRST STAGE REGRESSION DISCONTINUITY

Local linear regressions with triangular kernel x survey weights, heteroskedasticity-robust standard errors. Slope of running variable allowed to differ on either side of cutoff. Full set of control variables for Damages/Geo/Demo specifications includes self-reported earthquake, age and education of household head, a dummy for high caste, and the travel time to nearest health clinic. 'Close Point' refers to specifications where the running variable is distance to the specific point on the border that is closest to any village. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels respectively.

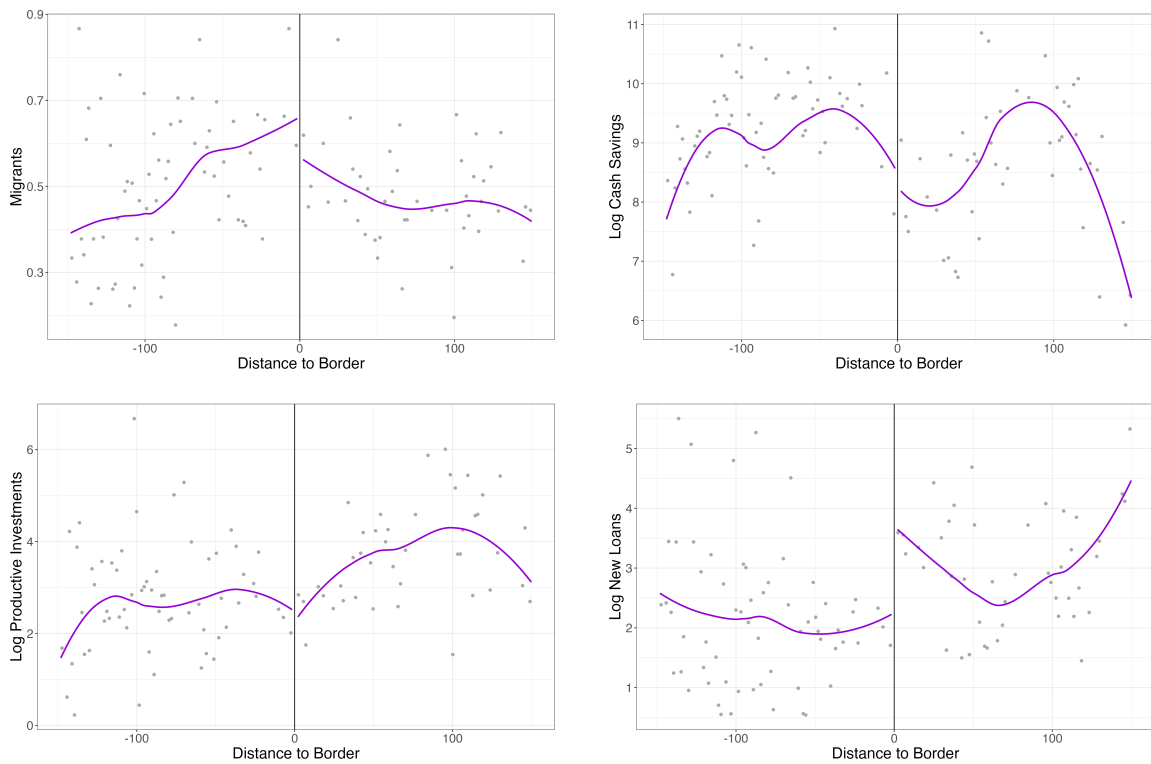


FIGURE B.3.—RD Plots: Binned averages of migrations, cash savings, productive investments and loans taken in the past year as a function of distance to the border with Loess smoothing. Positive values indicate villages inside the districts that were prioritized for aid.

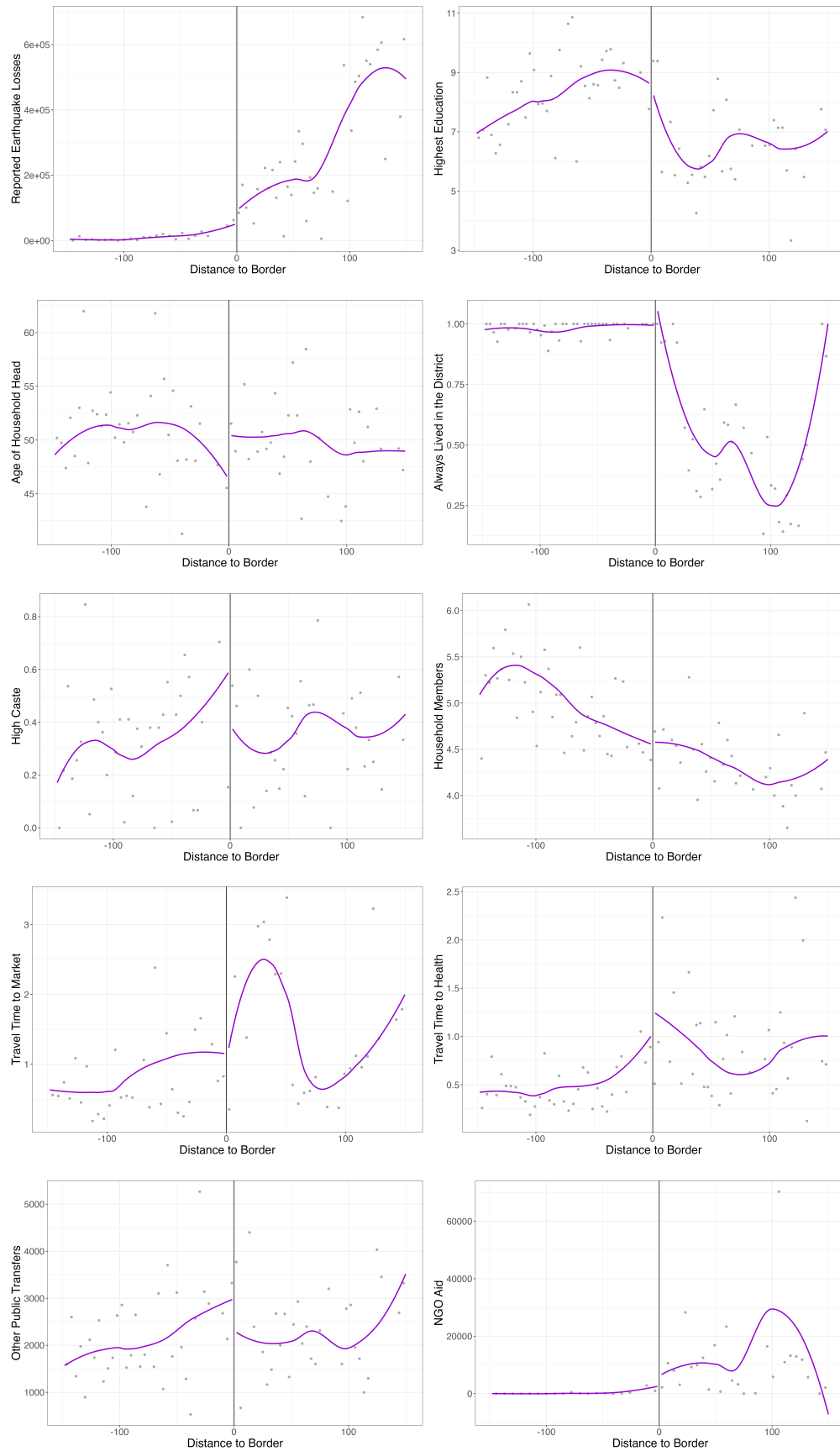


FIGURE B.4.—RD Plots: Binned averages of placebo variables from first survey wave as a function of distance to the border with Loess smoothing. Positive values indicate villages inside the districts that were prioritized for aid.

Running Variable: Distance to Border (Main Specification)											
	gorkha loss ever (1)	high caste (2)	age hh (3)	highest ed (4)	hhmembers (5)	class5 (6)	class10 (7)	always lived house (8)	always lived dist (9)	NGO transfers (10)	non quake aid (11)
$\widehat{Aid}$	222.734 (167.669)	-1.55 (1.44)	32.90 (33.93)	1.92 (5.59)	0.77 (2.61)	-0.01 (0.47)	0.96 (1.09)	0.53 (0.46)	0.53 (0.46)	17.545 (14.561)	-1.183 (9.049)
N	505	437	422	408	408	605	437	422	422	351	437
Bandwidth	44.05	41.72	39.06	37.41	37.39	50.75	41.23	39.18	39.18	34.38	39.62
Wards	57	60	60	62	48	64	55	62	62	41	61
	chicken price (1)	rice price (2)	lentil price (3)	sugar price (4)	mutton price (5)	time to school (6)	time to health (7)	time to market (8)	time to bank (9)	slope (10)	elevation (11)
$\widehat{Aid}$	-179 (203)	74.82 (81.53)	5.66 (21.86)	-10.56 (61.38)	-378 (620)	-0.02 (0.31)	-0.09 (0.82)	0.07 (1.51)	0.54 (1.52)	-16.72** (8.19)	318 (698)
N	171	262	205	300	48	437	251	309	251	194	422
Bandwidth	47.32	35.13	34.69	43.09	46.45	40.73	29.42	32.94	27.76	23.55	38.31
Wards	44	60	45	63	35	57	38	48	36	36	57
Running Variable: Distance to Single Point											
	gorkha loss ever (1)	high caste (2)	age hh (3)	highest ed (4)	hhmembers (5)	class5 (6)	class10 (7)	always lived house (8)	always lived dist (9)	NGO transfers (10)	non quake aid (11)
$\widehat{Aid}$	156,425* (94,240)	-0.22 (0.38)	19.88 (12.33)	4.63 (2.98)	0.03 (1.06)	-0.51 (0.36)	-0.10 (0.41)	0.26 (0.22)	0.26 (0.22)	14,609* (8,506)	-135 (3,686)
N	394	689	659	548	563	548	465	380	380	351	716
Bandwidth	43.73	62.9	61.13	55.29	55.89	54.76	49.05	42.57	42.57	42.16	63.91
Wards	59	91	85	77	77	82	71	57	57	60	91
	chicken price (1)	rice price (2)	lentil price (3)	sugar price (4)	mutton price (5)	time to school (6)	time to health (7)	time to market (8)	time to bank (9)	slope (10)	elevation (11)
$\widehat{Aid}$	-120 (96.17)	20.18 (14.66)	-32.60 (25.54)	12.49 (20.07)	-221 (230)	0.35** (0.15)	-0.05 (0.26)	-1.33** (0.59)	-1.23** (0.56)	-35.93** (14.63)	-947** (389)
N	278	300	194	442	53	659	604	154	125	336	308
Bandwidth	69.27	46.12	41.73	61.32	51.79	61.16	57.52	30.16	28.68	41.05	40.7
Wards	76	61	53	84	39	88	78	31	28	57	54

TABLE B.VII

PLACEBO TESTS:

Local linear regressions with triangular kernel times survey weights, heteroskedasticity-robust standard errors. Slope of running variable allowed to differ on either side of cutoff. Using data from first survey wave before aid disbursements. Aid instrumented with Distance > 0. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels respectively.

Control for Damages:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	0.69* (0.40)	10.48** (4.53)	-14.95*** (5.71)	0.76 (3.13)	7.29 (4.65)	2.87 (3.22)	1.95* (1.06)	-0.94** (0.44)
N	1259	1478	1572	1071	1440	1093	1223	1572
Bandwidth	37.54	43.78	45.64	34.76	42.62	34.48	37.14	45.22
Wards	66	66	66	64	66	66	66	66
Control for Damages, Demographics and Geography:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	0.70* (0.40)	10.60** (4.64)	-17.74** (7.17)	0.31 (3.08)	7.71 (5.09)	3.38 (3.40)	1.66 (1.02)	-1.12** (0.53)
N	1348	1565	1485	1245	1355	1310	1353	1529
Bandwidth	40.53	45.22	43.89	37.91	40.03	39.25	39.94	43.94
Wards	66	66	64	64	64	66	66	66
Dependent variable in Levels:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	111,707** (48,756)	334,950* (183,327)	-252,773** (109,834)	53,928 (106,629)	35,153 (60,519)	146,875 (104,180)	578,033** (251,357)	-0.76** (0.37)
N	777	955	917	898	1965	784	607	1093
Bandwidth	29.47	33.11	30.57	31.55	53.85	29.62	24.82	34.92
Wards	66	37	37	64	85	58	58	45
Dependent variable in Levels with Damage Controls:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	137,465** (68,506)	343,105 (232,673)	-357,456** (150,575)	69,048 (134,657)	81,507 (107,584)	185,569 (117,669)	959,000** (429,377)	-0.94** (0.44)
N	953	1086	1572	940	1355	1223	783	1572
Bandwidth	32.57	35.01	46.08	32.58	40.62	37.15	28.29	45.22
Wards	66	37	66	64	66	71	66	66
Dependent variable in Levels with Damage/Demographic/Geography Controls:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	112,297* (60,806)	428,120* (257,999)	-442,287** (195,891)	-50,304 (85,383)	32,985 (73,284)	25,721 (83,135)	903,055** (423,303)	-1.12** (0.53)
N	1084	1261	1485	2619	2399	2182	915	1529
Bandwidth	35.1	38.16	43.03	67.55	61.15	57.85	31.33	43.94
Wards	62	39	64	87	105	98	66	66

TABLE B.VIII

ROBUSTNESS CHECKS

Local linear regressions with triangular kernel times survey weights, heteroskedasticity-robust standard errors.

Slope of running variable allowed to differ on either side of cutoff. Full set of controls includes self-reported earthquake damages age and education of the household head, a dummy for high caste, a dummy for whether the household has always lived in the district, and the travel time to the nearest health clinic. Aid instrumented with

Distance > 0. Except where otherwise indicated, all dependent variables in logs, with 1 added to home investment, remittances, cash savings, loans, investments, and migration to account for zeros. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels respectively.

Alternate Bandwidths:							
	consumption	home investment	remittance income	cash savings	new loans taken	investments	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\widehat{Aid}$	0.62** (0.25)	2.03 (2.57)	-2.37 (2.89)	1.27 (2.02)	4.72* (2.81)	1.44 (2.16)	1.00 (0.65)
N	472	473	476	460	475	476	476
Bandwidth	20	20	20	20	20	20	20
Wards	11	11	11	11	11	11	11
$\widehat{Aid}$	0.68** (0.31)	6.30** (3.14)	-11.49** (4.50)	0.60 (2.41)	5.47 (3.48)	2.07 (2.51)	1.86** (0.85)
N	864	866	872	853	871	872	872
Bandwidth	30	30	30	30	30	30	30
Wards	21	21	21	21	21	21	21
$\widehat{Aid}$	0.58* (0.31)	9.22** (3.68)	-13.65*** (5.18)	0.69 (2.55)	6.51* (3.76)	1.18 (2.57)	1.73** (0.86)
N	1348	1350	1357	1334	1355	1355	1353
Bandwidth	40	40	40	40	40	40	40
Wards	32	32	32	32	32	32	32
$\widehat{Aid}$	0.52** (0.23)	9.55*** (2.85)	-10.51*** (3.50)	-0.36 (1.92)	5.48** (2.74)	0.24 (1.95)	1.21** (0.61)
N	1827	1829	1836	1808	1834	1834	1832
Bandwidth	50	50	50	50	50	50	50
Wards	44	44	44	44	44	44	44
$\widehat{Aid}$	0.22 (0.16)	8.61*** (2.23)	-8.10*** (2.58)	-1.78 (1.60)	4.92** (2.18)	-0.52 (1.59)	0.42 (0.46)
N	2349	2353	2360	2323	2355	2357	2355
Bandwidth	60	60	60	60	60	60	60
Wards	56	56	56	56	56	56	56

TABLE B.IX

ROBUSTNESS CHECKS: ALTERNATE BANDWIDTHS

Local linear regressions with triangular kernel times survey weights, heteroskedasticity-robust standard errors. Slope of running variable allowed to differ on either side of cutoff. Aid instrumented with Distance > 0. All dependent variables in logs, with 1 added to home investment, remittances, cash savings, loans, investments, and migration to account for zeros. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels respectively.

Uniform Kernel:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	0.42** (0.19)	4.44** (2.09)	-21.91** (9.17)	0.20 (1.62)	7.47 (4.70)	0.77 (1.69)	0.72 (0.49)	-0.65* (0.35)
N	472	473	739	460	871	476	476	608
Bandwidth	22.56	20.2	26.22	22.81	30.09	18.46	19.36	24.01
Wards	32	36	39	32	50	27	27	32
Epanechnikov Kernel:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	0.64** (0.31)	9.56** (3.92)	-2.53 (2.60)	0.24 (2.45)	6.81* (4.02)	1.91 (2.32)	0.98 (0.65)	-0.61* (0.32)
N	777	1130	476	766	1355	739	435	739
Bandwidth	28.79	36.14	19.54	28.47	41.52	26.38	17.54	26.6
Wards	58	56	30	47	73	50	31	37
5 km Donut Hole:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	0.04 (0.23)	3.43** (1.37)	-18.94*** (6.91)	-3.51*** (0.89)	1.39 (1.39)	-0.32 (1.24)	1.11 (0.81)	-0.69*** (0.21)
N	694	349	1228	335	391	392	1007	524
Bandwidth	28.04	17.06	38.33	16.82	22.7	20.59	35.68	24.67
Wards	60	28	57	29	30	29	85	35
Excluding Unaffected Districts:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	0.68** (0.32)	9.74*** (3.63)	-3.87 (2.80)	0.81 (2.15)	5.39 (3.43)	2.15 (2.56)	0.94 (0.61)	-0.65* (0.34)
N	953	1478	608	721	783	917	476	828
Bandwidth	32.87	43.07	23.92	26.13	29.21	31.38	21.67	29.82
Wards	59	63	32	37	37	63	34	39
Inverse Hyperbolic Sine of Dependent Variable:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	0.66** (0.32)	10.30*** (3.99)	-6.82** (3.39)	0.92 (2.70)	6.03 (3.78)	1.26 (2.74)	0.93 (0.61)	-0.76** (0.37)
N	1084	1435	695	984	916	1310	476	1093
Bandwidth	33.92	42.3	25.32	33.48	31.29	38.93	22.07	34.92
Wards	66	66	35	47	40	77	35	45

TABLE B.X

ROBUSTNESS CHECKS: ALTERNATE KERNELS AND OTHER

Local linear regressions with triangular kernel times survey weights, heteroskedasticity-robust standard errors. Slope of running variable allowed to differ on either side of cutoff. Aid instrumented with Distance > 0. Donut hole drops villages within 5km of either side of the border. Except where otherwise indicated, all dependent variables in logs, with 1 added to home investment, remittances, cash savings, loans, investments, and migration to account for zeros. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels respectively.

Distance to Single Point with Control Variables:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	0.11 (0.22)	3.82 (2.38)	-4.37 (2.74)	-0.60 (1.95)	4.59* (2.51)	0.10 (1.98)	0.35 (0.53)	-0.34 (0.23)
N	1039	2394	2010	1636	2140	1792	2396	2401
Bandwidth	41.44	67.11	60.77	51.88	62.92	56.87	67.38	67.71
Wards	55	99	85	69	88	78	99	98
Distance to Single Point with Dependent Variable in Levels:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	54,149* (29,105)	247,366** (105,611)	-86,271 (61,909)	14,457 (64,056)	15,552 (54,237)	-99,357 (110,729)	322,388** (159,599)	-0.38 (0.26)
N	344	1831	1705	1636	2311	2794	1043	1224
Bandwidth	27.52	57.1	54.92	51.77	66.18	80.23	41.97	42.74
Wards	32	78	85	84	80	85	66	57
Using Border of All 14 Districts:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	-0.03 (0.18)	1.04 (1.78)	-3.29 (2.44)	-5.27** (2.20)	-0.39 (1.83)	-4.91*** (1.83)	0.67 (0.61)	-0.19 (0.19)
N	2104	2194	2679	2873	2021	2153	2329	2108
Bandwidth	14.79	17.1	19.61	23.25	13.95	15.16	17.7	14.46
Wards	89	102	103	148	95	102	100	78
Using Border of All 14 Districts with Control Variables:								
	consumption	home investment	remittance income	cash savings	new loans taken	investments	total income	migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\widehat{Aid}$	0.11 (0.23)	-0.03 (2.05)	-1.31 (2.70)	1.05 (1.98)	-2.35 (2.43)	-5.58*** (2.25)	1.87** (0.85)	-0.42 (0.26)
N	1576	1843	2197	2176	2196	1801	2197	1669
Bandwidth	10.68	12.91	15.79	15.9	15.55	12.66	16.91	11.5
Wards	70	93	101	103	107	95	107	74

TABLE B.XI

ROBUSTNESS CHECKS: ALTERNATE RUNNING VARIABLE DEFINITIONS

Local linear regressions with triangular kernel times survey weights, heteroskedasticity-robust standard errors. Slope of running variable allowed to differ on either side of cutoff. Full set of controls includes self-reported earthquake damages, age and education of the household head, a dummy for high caste, a dummy for whether the household has always lived in the district, and the travel time to the nearest health clinic. Aid instrumented with Distance > 0. Single point is the point on the Western border of the 14 most-affected districts that is closest to any village. Except where otherwise indicated, all dependent variables in logs, with 1 added to home investment, remittances, cash savings, loans, investments, and migration to account for zeros. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels respectively.

APPENDIX C: APPENDIX C. DETERMINANTS OF EARTHQUAKE DAMAGES AND EFFECT OF EARTHQUAKE ON INCOME

A. All Survey Waves					
	(1)	(2)	log(income)		(5)
			(3)	(4)	
(Intercept)	11.752*** (0.009)	11.522*** (0.053)	11.717*** (0.012)	11.456*** (0.381)	14.876*** (1.286)
Earthquake Losses (000s NPR)	0.000 (0.000)	0.000 (0.000)	0.001*** (0.000)	0.020*** (0.008)	0.020*** (0.007)
log(elevation)					-0.310*** (0.110)
log(distance to epicenter)					-0.229 (0.148)
Observations	16 742	16 742	16 742	16 742	16 742
R2	0.000	0.043	-0.013	-4.636	-4.581
Fixed Effects		Village + Wave		District + Wave	District + Wave
IV			Shake Intensity	Shake Intensity	Shake Intensity

B. First Survey Wave					
	(1)	(2)	log(income)		(5)
			(3)	(4)	
(Intercept)	11.587*** (0.015)	11.566*** (0.085)	11.574*** (0.019)	11.770*** (0.498)	15.160*** (1.524)
Earthquake Losses (000s NPR)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.015 (0.009)	0.014* (0.007)
log(elevation)					-0.268** (0.124)
log(distance to epicenter)					-0.237 (0.181)
Observations	5547	5547	5547	5547	5547
R2	0.000	0.032	-0.002	-3.105	-2.361
Fixed Effects		Village		District	District
IV			Shake Intensity	Shake Intensity	Shake Intensity

TABLE C.I

EFFECTS OF EARTHQUAKE DAMAGES ON INCOME.

Panel A tests the effect of earthquake losses on log income using all three survey waves. Since households that experienced earthquake losses also were more likely to receive aid, this could include the effects of aid. Panel B tests the effect of earthquake losses on log income in the first survey wave before reconstruction aid disbursements began and finds fairly precise null effects, although the final two columns are positive and borderline significant. If anything this would bias estimates of WTP higher for households with higher earthquake damages. In both panels, column 1 is estimated using OLS. Column 2 adds village dummies and survey wave fixed effects in Panel A. Since higher losses could be associated with bigger houses and wealthier households, columns 3-5 instrument for earthquake losses using earthquake intensity (peak ground acceleration from USGS). Columns 4 and 5 add district dummies to control for regional differences in incomes, as well as wave fixed effects in panel A. Column 5 adds additional controls for elevation and distance to the epicenter. Earthquake losses self reported in thousands of NPR. Households with zero or missing income or less than 3 observations dropped. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

	Aid Eligible			
	(1)	(2)	(3)	(4)
Foundation:Cement/Stone/Brick	−0.084*** (0.003)			
Foundation:Mud mortar-Stone/Brick	0.162*** (0.002)			
Foundation:Other	0.041*** (0.007)			
Reinforced Concrete	−0.412*** (0.003)			
Roof:Bamboo/Timber-Light		0.018*** (0.001)		
Roof:Reinforced Concrete		−0.477*** (0.002)		
Total Square Feet			−0.0001*** (0.00000)	
Death or Injury occurred				0.056*** (0.003)
<i>N</i>	747,137	747,137	747,137	747,137
<i>R</i> ²	0.355	0.346	0.305	0.299

TABLE C.II

BUILDING MATERIALS

Estimated from post earthquake building census in the 11 most affected districts. Dependent variable is whether households were deemed eligible for aid, which indicates that engineers scored the damage to their home as a 4 or 5 on a 1-5 scale. Excluded category in column 1 is Bamboo/Timber foundation. Excluded category in column

2 is Heavy Bamboo/Timber roof. All regressions include village dummies. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels respectively.

	Aid Eligible				
	(1)	(2)	(3)	(4)	(5)
Owned TV	−0.064*** (0.001)				
Owned Mobile Phone		−0.040*** (0.001)			
Owned Land			0.074*** (0.002)		
Income: Rs. 10-20k NPR				−0.047*** (0.001)	
Income: Rs. 20-30k NPR				−0.085*** (0.002)	
Income: Rs. 30-50k NPR				−0.102*** (0.003)	
Income: Rs. 50k+ NPR				−0.118*** (0.005)	
Has Bank Account					−0.052*** (0.001)
<i>N</i>	747,137	747,137	747,137	747,137	747,137
<i>R</i> ²	0.302	0.300	0.300	0.303	0.301

TABLE C.III

HOUSEHOLD ASSETS AND INCOME

Estimated from post earthquake building census in the 11 most affected districts. Dependent variable is whether households were deemed eligible for aid, which indicates that engineers scored the damage to their home as a 4 or 5 on a 1-5 scale. All regressions include village dummies. Excluded category in column 4 is income under 10 thousand. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels respectively.

	Aid Eligible				
	(1)	(2)	(3)	(4)	(5)
Migrant Connected	−0.002** (0.001)				
Receives Social Security		0.026*** (0.001)			
Male Headed Household			0.036*** (0.001)		
Finished High School				−0.059*** (0.002)	
Some Middle/High School				−0.027*** (0.001)	
Newar					−0.054*** (0.002)
Other Caste					0.006*** (0.001)
<i>N</i>	747,137	747,137	747,137	747,137	747,137
<i>R</i> ²	0.299	0.299	0.300	0.300	0.300

TABLE C.IV

HOUSEHOLD DEMOGRAPHICS

Estimated from post earthquake building census in the 11 most affected districts. Dependent variable is whether households were deemed eligible for aid, which indicates that engineers scored the damage to their home as a 4 or 5 on a 1-5 scale. Excluded category in column 4 is less than high school. Excluded category in column 5 is high caste (Brahmin/Chhetri). All regressions include village dummies. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels respectively.

APPENDIX D: APPENDIX D. ALTERNATE COUNTERFACTUALS

Welfare Measure	Targeting Scenario						
	Actual (1)	Damages (2)	Consumption (3)	Liquidity (4)	Village Based (5)	Universal (6)	Optimal (7)
$\gamma = 1.2 \text{ and } \beta = .926$							
WTP	1.10 (1.05, 1.13)	1.10 (1.06, 1.13)	0.98 (0.93, 1.01)	0.82 (0.76, 0.84)	1.14 (1.09, 1.17)	1.01 (1.00, 1.03)	1.60 (1.54, 1.63)
Utilitarian	1.02 (0.97, 1.07)	1.01 (0.96, 1.05)	1.04 (0.99, 1.09)	0.94 (0.87, 0.98)	1.03 (0.97, 1.07)	1.00 (0.98, 1.02)	1.86 (1.81, 1.93)
$\bar{h} = 0.326$							
WTP	1.10 (1.05, 1.13)	1.10 (1.06, 1.13)	1.00 (0.95, 1.03)	1.01 (0.96, 1.04)	1.12 (1.07, 1.13)	1.12 (1.10, 1.13)	1.70 (1.62, 1.72)
Utilitarian	0.39 (0.10, 0.61)	0.46 (0.08, 0.71)	1.33 (0.65, 1.87)	2.48 (2.35, 2.65)	0.22 (0.04, 0.33)	0.98 (0.54, 1.41)	2.62 (2.51, 2.80)
$P_{it} \text{ based on pre-quake}$							
WTP	1.12 (1.07, 1.15)	1.13 (1.08, 1.16)	0.99 (0.95, 1.03)	0.96 (0.90, 0.99)	1.14 (1.09, 1.17)	1.11 (1.09, 1.12)	1.71 (1.63, 1.74)
$P_{it} \text{ based on post-aid}$							
WTP	1.09 (1.04, 1.12)	1.09 (1.04, 1.12)	0.99 (0.94, 1.03)	0.96 (0.91, 0.99)	1.12 (1.07, 1.15)	1.11 (1.09, 1.12)	1.71 (1.64, 1.74)

TABLE D.I

COUNTERFACTUAL TARGETING SCENARIOS

Benefits relative to a random allocation for 300,000 NPR aid allocated to 38% of the population according to various targeting strategies – except universal scenario which uses 0.38*300,000 NPR allocated to the entire population. Bootstrapped 95% confidence intervals in parentheses using survey weights. Utilitarian refers to an ‘equal-weighted’ utilitarian social welfare function. Consumption and liquidity allocations based on households with the lowest values of those variables. Damages based on households with the highest self-reported earthquake damages. Village based gives aid to all households in villages with the highest peak-ground acceleration from USGS.

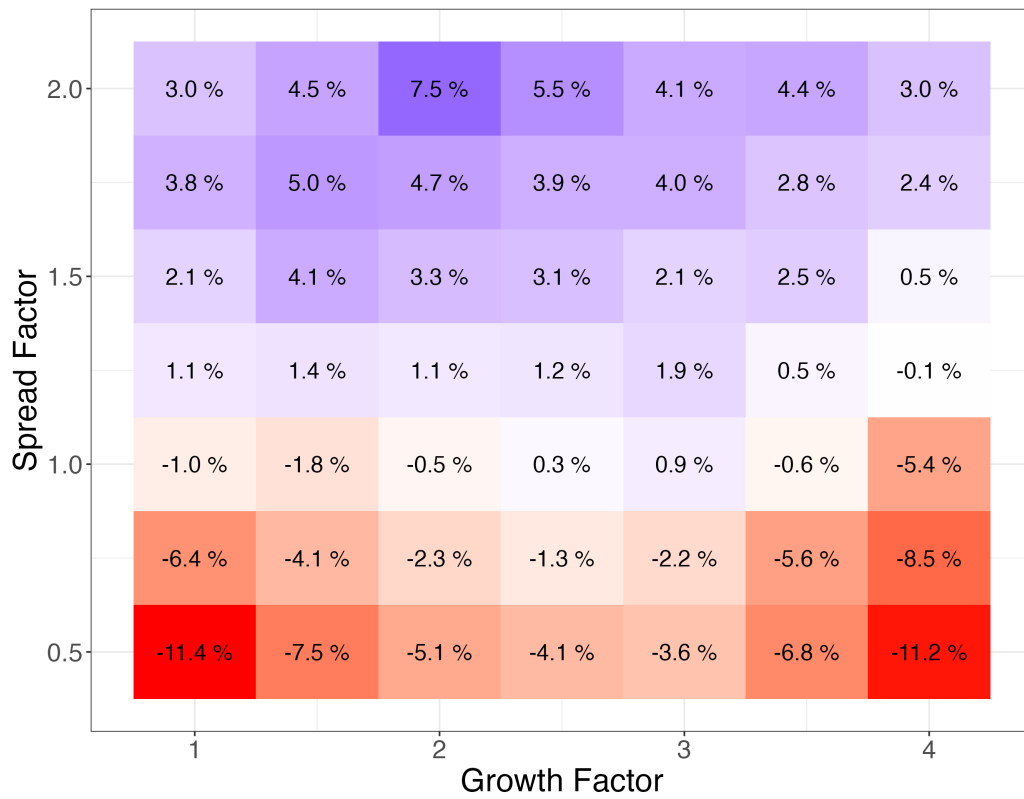


FIGURE D.1.—Alternate State Variable Distributions:
Aggregate Welfare Improvement from Targeting Relative to a Universal Allocation. Household state variables transformed as described in section 5.1. Targeting program considered is the actual distribution of aid recorded in the household survey.